
Know Thyself, Know Thy User: Intrinsic Dual-Perspective Reasoning for Role-Playing LLMs

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Abstract

As role-playing Large Language Models (LLMs) become central to personalized AI, they face a fundamental challenge: balancing character authenticity with user satisfaction. Prior dual-process and dual-perspective approaches address this through prompt-level conditioning, auxiliary modules, or inference-time reflection—realizing duality externally rather than within the core attention mechanism. We introduce the **KnowSelf-KnowOther Transformer (KSKT)**, which embeds dual-perspective reasoning directly into the generation process via dual-stream factorized attention that processes self-understanding and other-understanding through separate streams. This intrinsic integration enables token-level dynamic balance rather than post-hoc reconciliation. On CharacterBench, KSKT achieves 6.2% overall improvement over Qwen3-4B-Thinking. On SO-TOPIA, KSKT improves Relationship by 19.3% over the base model—the dimension requiring explicit self-other coordination. These results establish intrinsic dual-perspective reasoning as an effective architectural principle for role-playing systems.

1. Introduction

Consider a role-playing AI portraying a medieval peasant. When a user asks about calculus, the system faces an impossible choice: break character to provide helpful information, or maintain authenticity while leaving the user unsatisfied. This *character-user conflict*—the tension between preserv-

ing role authenticity and meeting user needs—is not an edge case but a fundamental challenge. As role-playing LLMs become central to personalized AI systems (Shao et al., 2023; Zhou et al., 2023; Tu et al., 2024), resolving this conflict has become increasingly critical: current systems typically exhibit single-perspective reasoning, either abandoning character to please users or rigidly adhering to role constraints while ignoring user intent.

Dual-process theories from cognitive science (Stanovich et al., 2000; Sloman, 1996; Evans, 2008) and their AI instantiations (Booch et al., 2021; Bellini-Leite, 2024) suggest that balancing competing perspectives requires coordinating fast intuitive responses with slow deliberative reasoning. Recent work has explored this direction through dual-model reflection frameworks (Li et al., 2025a), multi-agent debate mechanisms (Sun et al., 2023; Qian et al., 2024), and domain-specific dual-perspective architectures for tasks like metaphor detection (Lin et al., 2025) and multimodal reasoning (Ma et al., 2024). However, these approaches share a common limitation: they treat dual-perspective reasoning as an *external add-on*—something applied through prompt conditioning, separate modules, or cross-model coordination—rather than a capability *intrinsic* to the attention mechanism itself. This design choice leaves a structural gap: when a medieval peasant must respond to a calculus question, the optimal response requires weighing character constraints and user needs *at every token*. Unified attention can in principle learn to shift between perspectives implicitly; our claim is not that it cannot, but that *explicit* factorization supplies a stronger, more parameter-efficient and more interpretable inductive bias for this token-level arbitration—an inductive bias we validate empirically (§3.2, §4) rather than posit as a theoretical necessity.

We argue that dual-perspective reasoning must be **intrinsic**—embedded directly into the generation mechanism—to enable token-level dynamic balance between character authenticity and user satisfaction. To realize this principle, we introduce the **KnowSelf-KnowOther Transformer (KSKT)**, which extends standard transformer architectures with axial attention that processes self-understanding (character constraints) and other-understanding (user intentions) through separate streams with learnable fusion. Unlike ex-

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isting approaches where the base attention processes both perspectives uniformly, KSKT enables the model to simultaneously weigh “what would my character do?” and “what does the user need?” during each generation step. This architectural integration represents a paradigm shift: from external duality realization to intrinsic dual-perspective reasoning.

This intrinsic approach yields substantial improvements on benchmarks designed to probe character-user coordination. On CharacterBench (Zhou et al., 2025), KSKT achieves 6.2% overall improvement over Qwen3-4B-Thinking; at 4.1B parameters it surpasses all evaluated open-source models up to 72B (exceeding Qwen2-72B-Instruct with $\sim 18\times$ fewer parameters) and is competitive with frontier closed-source models, demonstrating that targeted architectural design can rival scale (full comparison in Appendix D.2). On SOTOPIA (Zhou et al., 2024), KSKT improves Relationship—the dimension requiring explicit self-other coordination—by **19.3%** over the base model. These gains appear precisely where our architectural hypothesis predicts: the largest improvements occur on metrics that explicitly require dual-perspective reasoning.

In detail, we make the following contributions:

1. **Intrinsic integration principle.** We provide evidence that embedding dual-perspective reasoning directly into the attention mechanism enables token-level balance between character authenticity and user satisfaction (§3.2).
2. **Architectural operationalization.** We show how dual-process principles can be realized through dual-stream factorized attention and adaptive deliberation allocation (§3.3).
3. **Empirical validation.** On SOTOPIA, KSKT improves Relationship by 19.3% over the base model—the largest gain among all dimensions. On CharacterBench, KSKT surpasses all evaluated open-source models up to 72B (exceeding Qwen2-72B-Instruct with $\sim 18\times$ fewer parameters) and is competitive with frontier closed-source models (§4).

2. Related Work

Our work integrates dual-process principles into transformer architecture for role-playing LLMs, addressing a key gap: most prior methods implement dual-perspective reasoning externally (prompting, auxiliary modules, or post-hoc reflection) rather than making it intrinsic to generation.

Role-Playing LLMs and the Character-User Conflict. Role-playing systems have evolved from static persona conditioning (Zhang et al., 2018; Dinan et al., 2019) to experience-based character learning (Shao et al., 2023;

Lu et al., 2024) and agentic architectures with memory, planning, and system-level components (Park et al., 2023; Huang et al., 2024). Training-time approaches focus on profile–dialogue alignment to reduce inconsistency (Yu et al., 2025). These advances improve character fidelity, but a fundamental tension remains: when character constraints conflict with user expectations, existing systems lack architectural mechanisms for arbitration. This gap is increasingly visible in evaluation: benchmarks have expanded from dialogue quality to character-specific assessment (Zhou et al., 2025; Tu et al., 2024; Wang et al., 2025), interactive settings (Light et al., 2023; Hua et al., 2023), and inner-thought probing (Xu et al., 2025)—revealing *when* role-playing fails without prescribing *how* to resolve role-user conflicts architecturally.

Existing Approaches to Dual-Perspective Reasoning.

Dual-process theories distinguish fast intuitive and slow deliberative reasoning (Stanovich et al., 2000; Bellini-Leite, 2024; Booch et al., 2021), inspiring LLM strategies from zero-shot prompting (Kojima et al., 2022) to role-based prompting (Kong et al., 2024). Recent work extends this through three main paradigms: (i) *prompt-level* approaches that inject perspective information into context but process it through unified attention; (ii) *module-level* approaches that add domain-specific components for dual-perspective processing (Lin et al., 2025; Ma et al., 2024; Qin et al., 2025); and (iii) *pipeline-level* approaches that separate generation and reflection across models or stages (Qian et al., 2024; Sun et al., 2023; Li et al., 2025a). Despite their diversity, these approaches share a common architectural property: the base attention mechanism processes both perspectives *uniformly*, without explicit factorization into separate reasoning streams.

Companion Work. A companion ICML 2026 paper by the same authors, BRIDGE (Jiang et al., 2026), addresses a complementary failure mode: *cross-turn* drift, where Observable–Latent–Memory states diverge silently over long horizons, via triangular fixed-point refinement with Lyapunov-bounded memory evolution. KSKT targets *within-turn* conflict—the instantaneous tension between character constraints and user intent. The two operate at non-overlapping timescales and compose naturally: BRIDGE’s memory states can condition KSKT’s per-token arbitration.

Key Distinction. Leveraging axial attention (Ho et al., 2019; Wang et al., 2020) and mixture-of-experts routing (Jacobs et al., 1991; Fedus et al., 2022; Jiang et al., 2024), we *factorize* attention into separate self-understanding and other-understanding streams. This architectural factorization—*informed* by work on self-knowledge in persona agents (Bhat et al., 2025) and the limitations of prompt-based conditioning (Kim et al., 2025)—enables explicit, token-level

arbitration during generation.

3. The KnowSelf-KnowOther Transformer

This section presents the KnowSelf-KnowOther Transformer (KSKT). We first formalize the problem setup (§3.1), then describe our three core components: Factorized Attention for maintaining separate self/other streams (§3.2), Adaptive Deliberation for conflict-aware reasoning depth (§3.3), and Expert Routing for dimension-specific specialization (§3.4). Finally, we describe the training protocol (§3.5). Figure 1 provides an architectural overview.

3.1. Problem Setup

Let the input to a role-playing system be $\mathbf{x} = (\mathbf{R}, \mathbf{U}, \mathcal{D})$, where $\mathbf{R} \in \mathbb{R}^{n_r \times d}$ denotes the role description (character traits, knowledge boundaries, behavioral constraints), $\mathbf{U} \in \mathbb{R}^{n_u \times d}$ denotes the user query (intent, emotional cues, task specifications), and $\mathcal{D}$ denotes the dialogue history. Here n_r, n_u are the number of role and user tokens respectively, and d is the hidden dimension. The goal is to generate a response $\mathbf{y}$ that satisfies both *role constraints* $\mathcal{C}(\mathbf{R})$ —responses must be permissible given the character’s identity and capabilities—and *user intent* $\mathcal{I}(\mathbf{U})$ —responses must be desirable given what the user seeks. The character-user conflict arises when these impose incompatible requirements: a medieval peasant asked about calculus cannot satisfy both role authenticity and user’s informational need simultaneously.

We build upon Qwen3-4B-Thinking-2507 (Yang et al., 2025), a thinking-only model that performs deliberation by default. Rather than toggling deliberation on/off, KSKT introduces *deliberation budget control*—determining how deeply to reason when perspectives conflict. Architecture compatibility details are provided in Appendix A.

Architectural Overview. KSKT addresses the character-user conflict through four integrated components (summarized in Algorithm 1): (1) *Dual-Stream Axial Attention* maintains $\mathbf{R}$ and $\mathbf{U}$ as separate representational streams with learnable fusion; (2) *Mutual-Understanding Position Encoding* augments positions with role/intent relevance signals; (3) *Bipolar Reasoning* allocates deliberation budget based on conflict complexity; (4) *Self-Awareness MoE* routes to dimension-specific experts (personality, knowledge, emotion, capability). These components enable token-level arbitration during generation rather than uniform processing of both perspectives.

3.2. Factorizing Self- and Other-Understanding

Standard attention computes a unified representation where $\mathbf{R}$ and $\mathbf{U}$ compete for the same representational capacity. We factorize attention into separate streams, borrowing the

factorization principle of axial attention (Ho et al., 2019; Wang et al., 2020)—here factorizing along the *self/other* perspective axes rather than spatial axes, so the mechanism may equivalently be read as *dual-perspective factorized attention*: a *self-stream* encoding role constraints $\mathcal{C}(\mathbf{R})$, and an *other-stream* encoding user intent $\mathcal{I}(\mathbf{U})$. We emphasize this is *perspective-specialized factorization*, not strict semantic separation; §4 (linear probing) shows the streams nonetheless acquire genuinely distinct, cross-dissociated representations. Let $\mathbf{H} \in \mathbb{R}^{n \times d}$ denote the hidden states at a given layer, where n is sequence length. Let $\mathbf{R}_{\text{proc}} \in \mathbb{R}^{n_r \times d}$ and $\mathbf{U}_{\text{proc}} \in \mathbb{R}^{n_u \times d}$ denote the processed role and user representations (preprocessing details in Appendix A.1).

Stream Decomposition. We factorize attention through separate projection matrices, allowing each stream to specialize. For hidden states $\mathbf{H}$:

$$\mathbf{Q}^{\text{self}}, \mathbf{K}^{\text{self}}, \mathbf{V}^{\text{self}} = \mathbf{H}\mathbf{W}_q^{\text{self}}, \mathbf{H}\mathbf{W}_k^{\text{self}}, \mathbf{H}\mathbf{W}_v^{\text{self}} \quad (1)$$

$$\mathbf{Q}^{\text{other}}, \mathbf{K}^{\text{other}}, \mathbf{V}^{\text{other}} = \mathbf{H}\mathbf{W}_q^{\text{other}}, \mathbf{H}\mathbf{W}_k^{\text{other}}, \mathbf{H}\mathbf{W}_v^{\text{other}} \quad (2)$$

where $\mathbf{W}_*^{\text{self}}, \mathbf{W}_*^{\text{other}} \in \mathbb{R}^{d \times d_k}$ are separate projections enabling perspective-specific specialization.

Biased Attention Computation. Each stream preferentially attends to its corresponding token types via learned attention biases:

$$\mathbf{A}^{\text{self}} = \text{softmax} \left(\frac{\mathbf{Q}^{\text{self}} \mathbf{K}^{\text{self}T}}{\sqrt{d_k}} + \mathbf{B}^{\text{self}} \right) \quad (3)$$

$$\mathbf{A}^{\text{other}} = \text{softmax} \left(\frac{\mathbf{Q}^{\text{other}} \mathbf{K}^{\text{other}T}}{\sqrt{d_k}} + \mathbf{B}^{\text{other}} \right) \quad (4)$$

where d_k is the per-head dimension (128 in our implementation, following the base model configuration), and $\mathbf{B}^{\text{self}}, \mathbf{B}^{\text{other}} \in \mathbb{R}^{n \times n}$ are learned biases emphasizing role-defining and user-relevant positions respectively.

Mutual-Understanding Position Encoding. Standard positional encoding treats all token distances uniformly, but dual-perspective reasoning depends on dialogue structure: role tokens remain critical regardless of distance, while recent user tokens carry heightened importance. Inspired by Contextual Position Encoding (Golovneva et al., 2024), we augment RoPE (Su et al., 2024) with learned relevance signals:

$$\mathbf{PE}_{\text{mutual}}(i) = \mathbf{PE}_{\text{base}}(i) + \mathbf{W}_{\text{self}} \mathbf{f}_{\text{self}}(i, \mathbf{R}_{\text{proc}}) + \mathbf{W}_{\text{other}} \mathbf{f}_{\text{other}}(i, \mathbf{U}_{\text{proc}}) \quad (5)$$

where $\mathbf{f}_{\text{self}}$ and $\mathbf{f}_{\text{other}}$ are lightweight MLPs encoding position i ’s relevance to character context and user intent respectively. Implementation details (injection timing, rotary interaction) are in Appendix A.2.

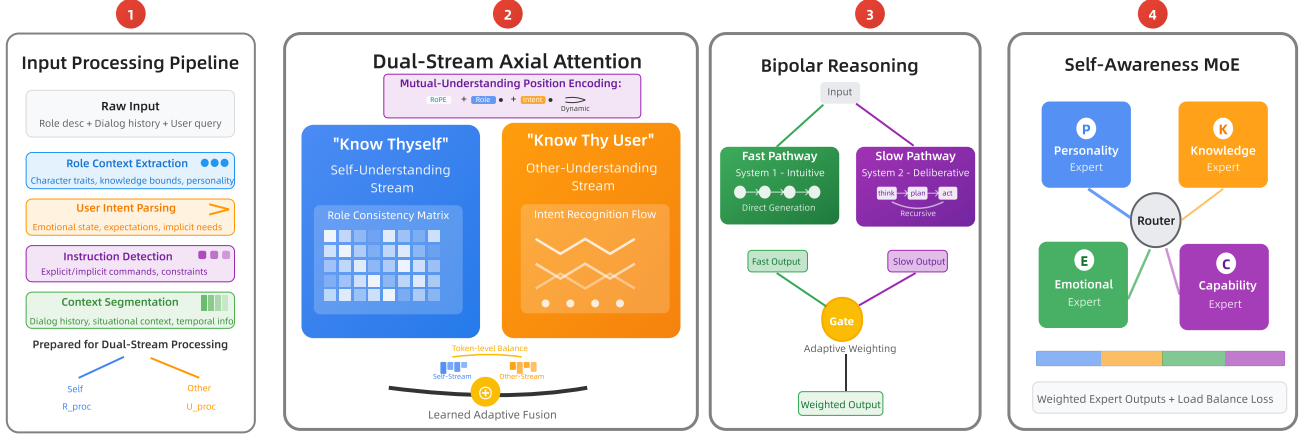


Figure 1. KSKT Architecture Overview. Our approach processes role-playing dialogues through four key components: (1) Input Processing Pipeline extracting role context and user intent; (2) Dual-Stream Axial Attention modeling self-understanding and other-understanding streams; (3) Bipolar Reasoning combining fast intuitive and slow deliberative pathways; and (4) Self-Awareness Mixture of Experts with personality (P), knowledge (K), emotional (E), and capability (C) specialists.

Adaptive Fusion. Having maintained both perspectives simultaneously, the model must decide *at each token* how to balance them. We achieve this through learned, token-level fusion weights:

$$\begin{aligned} \mathbf{s} &= \mathbf{H}\mathbf{w} + b \\ \alpha &= \sigma(\mathbf{s}), \quad \beta = \mathbf{1}_n - \alpha \end{aligned} \quad (6)$$

where $\mathbf{w} \in \mathbb{R}^d$ and $b \in \mathbb{R}$ are learnable parameters, producing $\alpha, \beta \in \mathbb{R}^{n \times 1}$ that strictly satisfy $\alpha + \beta = \mathbf{1}_n$.

$$\mathbf{H}^{\text{out}} = \alpha \odot (\mathbf{A}^{\text{self}} \mathbf{V}^{\text{self}}) + \beta \odot (\mathbf{A}^{\text{other}} \mathbf{V}^{\text{other}}) \quad (7)$$

where token-level weights $\alpha, \beta \in \mathbb{R}^{n \times 1}$ are broadcast across the hidden dimension d , providing interpretable perspective balancing at each position. For instance, when a medieval peasant character is asked about calculus, α -dominant tokens maintain character voice while β -dominant tokens address user needs—a dynamic visualized in detail in Section 4.5.

Simplex Constraint. Setting $\beta = \mathbf{1}_n - \alpha$ enforces $\alpha + \beta = \mathbf{1}_n$ with $\alpha, \beta \geq 0$, making fusion a strict trade-off rather than signal accumulation. This mirrors MoE gating (Jacobs et al., 1991), with the added benefit that α is directly interpretable as self-perspective trust.

3.3. Adaptive Deliberation for Perspective Arbitration

Not all generation steps require equal deliberation: aligned perspectives need minimal reasoning, while conflicts require explicit coordination. Our Bipolar Reasoning Module integrates fast intuitive (System 1) and slow deliberative (System 2) processing, controlling deliberation depth rather than toggling thinking on/off.

Dual Pathways. The key distinction between pathways is the depth of perspective coordination:

Let $\mathbf{H}^{\text{DSAA}}$ denote the output of Dual-Stream Axial Attention (§3.2). *System 1* (low-deliberation) produces responses through a single feedforward transformation, suitable when perspectives are already aligned:

$$\mathbf{H}^{\text{fast}} = \text{FFN}_{\text{fast}}(\text{RMSNorm}(\mathbf{H}^{\text{DSAA}})) \quad (8)$$

System 2 (extended-deliberation) allocates additional reasoning steps through a controlled thinking chain:

$$\mathbf{H}^{\text{slow}} = \text{ThinkingChain}(\mathbf{H}^{\text{DSAA}}, T) \quad (9)$$

where $T \in \{2, 4, 6, 8, 10\}$ is the thinking budget predicted by a two-layer MLP classifier. Each step $\mathbf{t}_i$ attends to both role and user context via cross-attention:

$$\mathbf{t}_i = \text{MLP}\left(\text{concat}(\mathbf{h}_{i-1}, \text{CrossAttention}(\mathbf{h}_{i-1}, [\mathbf{R}_{\text{proc}}; \mathbf{U}_{\text{proc}}]))\right) \quad (10)$$

where $\mathbf{h}_0 = \mathbf{H}^{\text{DSAA}}$ and $\mathbf{h}_i = \mathbf{h}_{i-1} + \mathbf{t}_i$ follows residual connection. During training, T is supervised using heuristic conflict scores derived from (i) fusion weight divergence $|\alpha - \beta|$ and (ii) expert routing entropy $\mathcal{S}(p_{\text{expert}})$, which serve as proxies for conflict complexity. The classifier is trained jointly with the main objective using cross-entropy loss (details in Appendix A.3).

Integration. The fast pathway (System-1) is computed for all tokens. A lightweight pre-gate, computed solely from pre-arbitration states $(\mathbf{H}^{\text{DSAA}}, \mathbf{H}^{\text{fast}})$, determines whether to invoke the System-2 thinking chain. When System-2 is triggered (approximately 25% of cases based on conflict scores), a post-gate fuses the fast and slow representations. The budget T controls *reasoning depth* only when System-2 is activated, yielding an expected overhead of $\sim 19.5\%$

rather than worst-case always-on cost (Appendix F):

$$\mathbf{H}^{\text{reason}} = \mathbf{g} \odot \mathbf{H}^{\text{fast}} + (1 - \mathbf{g}) \odot \mathbf{H}^{\text{slow}} \quad (11)$$

$$\mathbf{g} = \sigma(\mathbf{w}_g^T [\mathbf{H}^{\text{DSAA}}; \mathbf{H}^{\text{fast}}; \mathbf{H}^{\text{slow}}] + b_g) \in \mathbb{R}^{n \times 1} \quad (12)$$

where $\mathbf{w}_g \in \mathbb{R}^{3d}$ produces a token-level scalar gate, and $\mathbf{H}^{\text{reason}}$ denotes the post-arbitration hidden states that are subsequently routed through the Self-Awareness MoE (§3.4). Budget control remains valuable when System-2 is activated: (i) limiting chain length reduces reasoning drift and output noise, (ii) shorter chains improve training stability by preventing gradient degradation, and (iii) the learned budget provides interpretable conflict-complexity signals for analysis.

3.4. Multi-Dimensional Character Specialization

The preceding mechanisms operate uniformly across character dimensions. However, character-user conflicts manifest differently across dimensions: a character may have sophisticated *emotional* awareness but limited *domain knowledge*, or strong *personality* constraints but flexible *capability* boundaries. Uniform processing cannot capture these dimension-specific dynamics. We address this through specialized routing to dimension-specific experts.

Dimension Selection Rationale. We factorize character state into four axes—*Personality*, *Knowledge*, *Emotion*, and *Capability*—grounded in cognitive modeling (Big Five (McCrae & John, 1992), OCC (Ortony et al., 2022), BDI (Rao et al., 1995)). This covers the four core state variables: who the character is, what they know, how they feel, and what they can do.

Expert Specialization. Four expert networks $\{E_P, E_K, E_E, E_C\}$ handle personality, knowledge, emotion, and capability respectively. Each uses SwiGLU activation consistent with Qwen3:

$$E_j(\mathbf{h}) = \text{SwiGLU}(\mathbf{W}_1^{(j)} \mathbf{h} + \mathbf{b}_1^{(j)}) \mathbf{W}_2^{(j)} + \mathbf{b}_2^{(j)} \quad (13)$$

Self-Reflective Routing. Unlike standard MoE that routes based solely on input, our routing conditions on *which character dimension is most relevant*—e.g., prioritizing the Emotional Expert for responses to criticism, or the Knowledge Expert for domain boundary explanations. We extract character-specific routing signals by attending to the role context:

$$\mathbf{q}_{\text{self}} = \text{SwiGLU}(\mathbf{W}_{\text{concat}} [\mathbf{h}_{\text{role}}; \mathbf{h}_{\text{context}}]) \quad (14)$$

where $\mathbf{h}_{\text{role}}$ aggregates character-relevant signals from the post-arbitration hidden states $\mathbf{H}^{\text{reason}}$ via attention over $\mathbf{R}_{\text{proc}}$, and $\mathbf{h}_{\text{context}} = \text{MeanPool}(\mathbf{R}_{\text{proc}})$ provides global role

context (full formulation in Appendix A.4). Routing probabilities are then computed as:

$$p_j = \frac{\exp((\mathbf{w}_j^T \mathbf{q}_{\text{self}} + b_j)/\tau)}{\sum_{k \in \{P, K, E, C\}} \exp((\mathbf{w}_k^T \mathbf{q}_{\text{self}} + b_k)/\tau)}, \quad \tau > 0 \quad (15)$$

where τ controls distribution sharpness. Detailed formulations are in Appendix A.4.

Expert Combination. Weighted combination with load balancing:

$$\mathbf{H}^{\text{expert}} = \sum_{j \in \{P, K, E, C\}} p_j \cdot E_j(\mathbf{H}^{\text{reason}}) \quad (16)$$

$$\mathcal{L}_{\text{balance}} = \sum_j (f_j - 0.25)^2$$

where f_j is the fraction of tokens routed to expert j . The balance loss is weighted by $\lambda_3 = 0.01$ in the total objective (Equation (17)), preventing expert collapse while preserving specialization (Appendix A.4).

3.5. Learning Dual-Perspective Reasoning

Algorithm 1 KSKT Forward Pass

Input: Role description $\mathbf{R}$, user query $\mathbf{U}$, dialogue history $\mathcal{H}$

Output: Output hidden states $\mathbf{H}$

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1:  $\mathbf{R}_{\text{proc}}, \mathbf{U}_{\text{proc}} \leftarrow \text{PREPROCESS}(\mathbf{R}, \mathbf{U}, \mathcal{H})$ 
2:  $\mathbf{H} \leftarrow \text{EMBED}([\mathbf{R}; \mathbf{U}; \mathcal{H}])$ 
3: for each KSKT layer  $\ell$  do
4:   /* Dual-Stream Axial Attention */
5:    $\mathbf{H}^{\text{self}} \leftarrow \text{SELFATTN}(\mathbf{H}, \mathbf{R}_{\text{proc}})$ 
6:    $\mathbf{H}^{\text{other}} \leftarrow \text{OTHERATTN}(\mathbf{H}, \mathbf{U}_{\text{proc}})$ 
7:    $\alpha, \beta \leftarrow \text{FUSE}(\mathbf{H})$   $\triangleright \alpha + \beta = 1$ 
8:    $\mathbf{H}^{\text{dsaa}} \leftarrow \alpha \odot \mathbf{H}^{\text{self}} + \beta \odot \mathbf{H}^{\text{other}}$   $\triangleright \text{Eq. (7)}$ 
9:   /* Bipolar Reasoning */
10:   $\mathbf{H}^{\text{fast}} \leftarrow \text{FASTPATH}(\mathbf{H}^{\text{dsaa}})$ 
11:  if  $\text{PREGATE}(\mathbf{H}^{\text{dsaa}}, \mathbf{H}^{\text{fast}}) > \tau$  then
12:     $T \leftarrow \text{BUDGET}(\mathbf{H}^{\text{dsaa}})$ 
13:     $\mathbf{H}^{\text{slow}} \leftarrow \text{THINK}(\mathbf{H}^{\text{dsaa}}, T)$ 
14:     $g \leftarrow \text{POSTGATE}(\mathbf{H}^{\text{dsaa}}, \mathbf{H}^{\text{fast}}, \mathbf{H}^{\text{slow}})$ 
15:     $\mathbf{H}^{\text{reason}} \leftarrow g \odot \mathbf{H}^{\text{fast}} + (1-g) \odot \mathbf{H}^{\text{slow}}$   $\triangleright \text{Eq. (11)}$ 
16:  else
17:     $\mathbf{H}^{\text{reason}} \leftarrow \mathbf{H}^{\text{fast}}$ 
18:  end if
19:  /* Self-Awareness MoE */
20:   $\mathbf{p} \leftarrow \text{ROUTER}(\mathbf{H}^{\text{reason}}, \mathbf{R}_{\text{proc}})$   $\triangleright \text{Eq. (15)}$ 
21:   $\mathbf{H} \leftarrow \sum_{j \in \{P, K, E, C\}} p_j \cdot E_j(\mathbf{H}^{\text{reason}})$ 
22: end for
23: return  $\mathbf{H}$ 
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Training Protocol. Our training preserves the base model’s native deliberation behavior while learning two additional

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MC: Memory Consistency FA: Fact Accuracy BC _K : Boundary Consistency AC ^b : Attribute Consistency (Bot) AC ^h : Attribute Consistency (Human) BC _P ^b : Behavior Consistency (Bot) BC _P ^h : Behavior Consistency (Human) ES: Emotional Self-regulation ER: Empathetic Responsiveness MS: Morality Stability MR: Morality Robustness HL: Human-likeness EG: Engagement														
Models	AVG. zh/en	Memory	Knowledge		Persona				Emotion		Morality		Believability	
		MC zh/en	FA zh/en	BC _K zh/en	AC ^b zh/en	AC ^h zh/en	BC _P ^b zh/en	BC _P ^h zh/en	ES zh/en	ER zh/en	MS zh/en	MR zh/en	HL zh/en	EG zh/en
Closed-sourced LLMs														
GPT-3.5-turbo	3.66/3.72	3.83/3.58	2.43/2.52	3.57/3.75	4.33/4.38	4.13/4.23	3.37/3.50	3.51/3.58	3.07/3.14	2.85/2.81	4.76/4.71	4.84/4.71	3.32/3.69	3.54/3.74
GPT-4-1106	3.69/3.74	3.97/3.88	2.85/2.71	3.73/4.03	4.42/4.52	4.14/4.10	3.35/3.59	3.37/3.43	3.07/3.09	2.96/2.95	4.81/4.74	4.76/4.72	3.21/3.34	3.32/3.50
Claude-3-opus	3.83/3.88	3.98/4.01	2.69/2.50	4.10/4.45	4.57/4.54	4.39/4.44	3.72/3.74	3.73/3.77	3.45/3.63	3.15/3.15	4.88/4.91	4.80/4.68	2.95/3.23	3.34/3.44
Open-sourced LLMs														
CharacterGLM-6B	3.21/3.21	3.31/3.22	2.26/2.01	3.22/3.60	3.19/3.28	3.44/3.49	3.05/3.01	3.01/2.90	2.80/2.84	2.55/2.51	4.58/4.51	4.64/4.78	2.70/2.64	2.95/2.98
GLM4-9B-Chat	3.58/3.59	3.80/3.49	2.65/2.21	3.42/3.59	4.12/4.41	3.94/4.10	3.29/3.28	3.47/3.52	2.96/2.99	2.99/2.87	4.77/4.69	4.72/4.65	3.04/3.32	3.36/3.49
Llama3-8B-Instruct	3.60/3.65	3.98/3.72	2.35/2.35	3.49/3.81	4.42/4.29	4.26/4.27	3.51/3.57	3.32/3.50	3.04/3.14	2.93/3.07	4.84/4.81	4.80/4.76	2.69/2.99	3.12/3.23
Qwen2-7B-Instruct	3.66/3.51	4.18/3.86	2.76/2.27	3.45/3.66	4.46/4.51	4.07/3.91	3.47/3.23	3.31/3.18	3.11/2.96	3.12/2.85	4.88/4.73	4.91/4.74	2.76/2.78	3.06/2.96
Qwen2-72B-Instruct	3.80/3.68	4.03/3.94	3.00/2.59	3.85/3.95	4.53/4.39	4.22/3.96	3.53/3.33	3.35/3.35	3.25/3.06	3.14/2.89	4.92/4.71	4.85/4.74	3.30/3.40	3.41/3.51
Proposed Models														
Qwen3-4B-Thinking (Base)	3.69/3.63	4.05/3.85	2.65/2.30	3.65/3.75	4.35/4.30	4.15/4.10	3.65/3.60	3.55/3.50	3.10/2.95	2.95/2.82	4.80/4.75	4.75/4.70	3.10/3.20	3.20/3.30
+ FT (Same Data/Recipe)	3.80/3.75	4.12/4.00	2.69/2.33	3.72/3.82	4.52/4.47	4.31/4.26	3.81/3.75	3.71/3.65	3.29/3.13	3.11/2.97	4.83/4.78	4.78/4.73	3.23/3.38	3.33/3.47
+ Dual-Stream Axial Attention	3.78/3.72	4.18/3.98	2.76/2.37	3.79/3.95	4.45/4.38	4.21/4.05	3.48/3.28	3.35/3.37	3.55/3.55	3.28/3.30	4.81/4.65	4.75/4.68	3.20/3.33	3.33/3.47
+ Mutual-Understanding Position Encoding	3.80/3.74	4.20/4.00	2.78/2.39	3.80/3.96	4.47/4.40	4.23/4.07	3.50/3.30	3.37/3.39	3.58/3.58	3.30/3.32	4.84/4.68	4.78/4.71	3.21/3.34	3.34/3.48
+ Bipolar Reasoning Module	3.76/3.68	4.17/3.97	2.82/2.45	3.84/4.00	4.44/4.36	4.20/4.03	3.45/3.25	3.33/3.34	3.25/3.15	3.08/2.95	4.86/4.70	4.80/4.74	3.27/3.36	3.40/3.52
+ Self-Awareness MoE	3.79/3.73	4.00/3.81	2.70/2.35	3.74/3.86	4.68/4.61	4.51/4.52	3.83/3.81	3.81/3.84	3.18/3.02	3.05/2.85	4.79/4.63	4.73/4.66	3.06/3.21	3.19/3.32
KSKT (Full)	3.92/3.87	4.17/4.10	2.70/2.35	3.74/3.84	4.73/4.67	4.51/4.46	3.97/3.91	3.86/3.80	3.57/3.40	3.40/3.25	4.85/4.80	4.80/4.75	3.28/3.44	3.38/3.54

Table 1. Performance comparison on CharacterBench following the standardized evaluation protocol (Zhou et al., 2025). + FT (Same Data/Recipe) uses identical data and training recipe as KSKT. Each “+” row independently adds one component to Qwen3-4B-Thinking (Base); **KSKT (Full)** combines all. **Bold**: best in column.

capabilities: (i) *deliberation budget allocation*—how deeply to reason on each conflict, and (ii) *perspective capacity allocation*—how to balance self-understanding and other-understanding streams. Training uses a single multi-objective loss with *fixed* weights:

$$\mathcal{L} = \mathcal{L}_{\text{CLM}} + \lambda_1 \mathcal{L}_{\text{consistency}} + \lambda_2 \mathcal{L}_{\text{understanding}} + \lambda_3 \mathcal{L}_{\text{balance}} \quad (17)$$

where auxiliary losses encourage character identity preservation ($\mathcal{L}_{\text{consistency}}$), user intent comprehension ($\mathcal{L}_{\text{understanding}}$), and expert load balancing ($\mathcal{L}_{\text{balance}}$). We set $\lambda_2 > \lambda_1 > \lambda_3$, reflecting the relative importance of user-intent modeling; exact values are in Appendix B.2.

Principled Training Schedule. Optimizing all objectives from the start causes gradient interference: the model cannot establish stable character representations while simultaneously learning user-intent modeling. We address this through *phase-wise loss gating*: since effective user modeling requires stable character representations as a reference point, we progressively activate auxiliary losses rather than optimizing all objectives simultaneously.

Concretely, training follows three phases:

- **Phase 1: Self-Understanding Foundation** (2 epochs): $\mathcal{L}_{\text{CLM}}$ and $\mathcal{L}_{\text{consistency}}$ active. The model establishes stable character representations.
- **Phase 2: Other-Understanding Addition** (1 epoch): $\mathcal{L}_{\text{understanding}}$ additionally enabled. The model learns user-intent modeling atop the character foundation.
- **Phase 3: Mutual Integration** (1 epoch): $\mathcal{L}_{\text{balance}}$ additionally enabled. The model learns to coordinate both streams through adaptive fusion.

Note that $\mathcal{L}_{\text{CLM}}$ remains active throughout all phases to prevent catastrophic forgetting. All phases operate on the same training mixture, differing only in which auxiliary losses are activated. Implementation details are provided in Appendix A.

Summary. Having described each component, we summarize the complete KSKT forward pass in Algorithm 1, which integrates dual-stream attention (§3.2), adaptive deliberation (§3.3), and self-awareness routing (§3.4) into a unified generation procedure.

4. Experiments

4.1. Experimental Setup

Training Data and Protocol. We construct 130K training samples from three sources: 15K synthesized character profiles (Ge et al., 2024), 55K general instruction-following demonstrations (Taori et al., 2023; Zheng et al., 2023), and 60K role-specific dialogues. The general instruction data (42%) serves as a regularizer to prevent catastrophic forgetting (Luo et al., 2025). To disentangle data effects from architectural contributions, we additionally fine-tune the base model on the same mixture with an identical recipe (Qwen3-4B-Thinking + FT). Training initializes from Qwen3-4B-Thinking-2507 (Yang et al., 2025) using three progressive phases as described in §3.5. Complete hyperparameters and contamination prevention details are provided in Appendix B.2.

Evaluation Protocol. We evaluate on CharacterBench (Zhou et al., 2025), covering 13 metrics across six dimensions on 5-point scales in Chinese/English: *Memory* (dialogue recall), *Knowledge* (factual accuracy and boundaries), *Persona* (attribute and behavior consistency), *Emotion* (self-

regulation and empathy), *Morality* (ethical adherence), and *Believability* (human-likeness and engagement). Results report mean performance across three runs. Extended results, dynamic activation analysis, and comparisons with closed-source models appear in Appendices D–D.5.

4.2. Main Results

Table 1 presents results on CharacterBench. We analyze (i) architecture vs. data, (ii) efficiency vs. scale, and (iii) where gains concentrate.

Architectural contribution beyond data. With identical data and recipe, +FT improves 3.69→3.80 (+0.11), while KSKT further improves to 3.92 (+0.12 over +FT). We now isolate how much of this +0.12 is architectural rather than an artifact of richer supervision.

Isolating architecture from supervision. A natural concern is that KSKT’s auxiliary losses ($\mathcal{L}_{\text{con}}$, $\mathcal{L}_{\text{und}}$, $\mathcal{L}_{\text{balance}}$)—rather than the dual-stream factorization—drive the gain. These losses are, however, undefined on a standard architecture: $\mathcal{L}_{\text{con}}/\mathcal{L}_{\text{und}}$ require per-stream outputs and $\mathcal{L}_{\text{balance}}$ requires expert routing. To disentangle the two under *iso-parameter* ($\sim 4.1\text{B}$) and *iso-supervision* conditions, we build a deliberately favorable control—unified attention with doubled QKV projections plus proxy losses approximating our objectives, trained with the identical three-phase schedule, data, and loss weights (Wider-FT + Proxy Losses). As Table 2 shows, supervision enrichment alone yields only +0.03 ($\sim 25\%$), while the explicit factorization accounts for the remaining +0.09 ($\sim 75\%$, $p < 0.001$). The architectural gain concentrates in *Persona* ($\Delta\mathbb{C}^b$: +0.16) and *Emotion* ($\Delta\mathbb{E}$: +0.24) with Knowledge/Morality ≤ 0.01 ($p > 0.3$)—a dimension-specific pattern that generic supervision enrichment cannot explain.

Configuration	AVG (zh)	Δ from +FT	Attribution
+FT (baseline)	3.80	—	—
Wider-FT + Proxy Losses	3.83	+0.03 ($p < 0.05$)	$\sim 25\%$ sup.
KSKT (Full)	3.92	+0.12	—
Gap: Wider-FT→KSKT	—	+0.09 ($p < 0.001$)	$\sim 75\%$ arch.

Table 2. Iso-parameter, iso-supervision attribution. The explicit dual-stream factorization accounts for $\sim 75\%$ of KSKT’s gain over a data- and supervision-matched standard architecture.

Architectural efficiency vs. parameter scaling. At 4.1B parameters, KSKT exceeds Qwen2-72B-Instruct on both Chinese (3.92 vs. 3.80) and English (3.87 vs. 3.68) with $\sim 18\times$ fewer parameters, and surpasses every open-source baseline in Table 1 despite its far smaller scale. A broader comparison against newer frontier models—both open- and closed-source—is deferred to Appendix D.2 (Table 9), where KSKT remains competitive with the strongest closed-source systems; the small residual English gap likely reflects base-model biases (Appendix E). We next localize

the gains by dimension.

Dimension-specific improvements. Gains concentrate in dimensions aligned with our design goals. *Persona* improves most (Attribute Consistency: 4.73 vs. 4.35; Behavior Consistency: 3.97 vs. 3.65), supporting SAMOE’s character-centric specialization. Empathetic Responsiveness (ER) also improves substantially ($2.95 \rightarrow 3.40$, +15.3%), consistent with DSAA’s contribution to other-understanding. Moderate *Knowledge* reflects a deliberate trade-off: we prioritize role authenticity and boundary-keeping over encyclopedic breadth.

These dimension-specific patterns suggest that each architectural component contributes distinctly. We examine this hypothesis through systematic ablation in the following section. Significance tests ($p < 0.001$, Cohen’s $d = 0.76$ – 1.15) and human evaluation are reported in Appendices C and D.7.

Efficiency-performance trade-off. KSKT introduces 19.5% computational overhead (Appendix F), comparable to standard MoE architectures (Jiang et al., 2024). Critically, the architectural contribution (+0.12 from FT baseline to KSKT Full) roughly equals the data contribution (+0.11 from Base to FT), demonstrating genuine architectural value beyond data effects. Moreover, KSKT’s modular design enables selective component activation (Appendix A.5), where simple queries utilize minimal overhead while complex conflicts engage the full architecture.

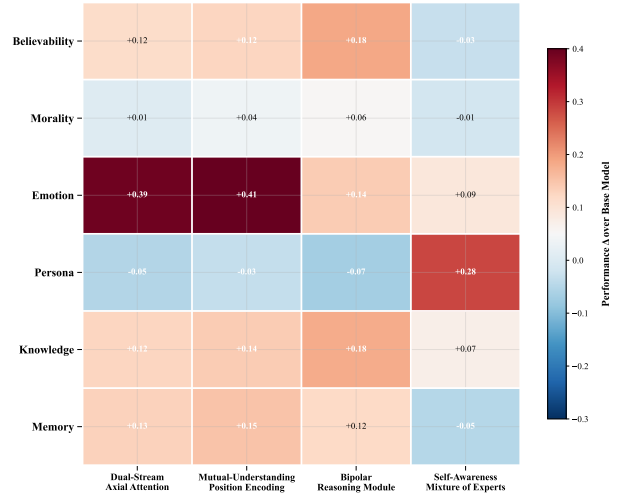


Figure 2. Component specialization on CharacterBench: delta over Qwen3-4B-Thinking aggregated into six dimensions. Cell values are exact deltas (higher is better).

4.3. Architectural Component Analysis

Figure 2 shows one-at-a-time ablations. Components targeting *other-understanding*—DSAA and MUPE—produce the strongest gains on Emotion (+0.39 and +0.41), which mea-

sures empathetic responsiveness to users. Conversely, the component targeting *self-understanding*—Self-Awareness Mixture of Experts (SAMOE)—delivers dominant Persona improvement (+0.28) while slightly trading off Memory (−0.05).

Bipolar Reasoning Module (BRM) exhibits a qualitatively different profile: balanced contributions across Believability (+0.18) and Knowledge (+0.18), consistent with its role as a deliberation coordinator rather than a dimension-specific specialist. The complementary specializations—DSAA/MUPE for other-understanding, SAMOE for self-understanding, BRM for coordination—motivate evaluation on interactive scenarios where both perspectives must be dynamically balanced.

Do the streams genuinely specialize? To verify that the two streams encode distinct information—rather than redundant projections of the same signal—we train single-layer linear classifiers on the *frozen* self- and other-stream outputs of the final DSAA layer (SGD, 50 epochs, 3 seeds). Table 3 shows a clean *cross-dissociation*: the self-stream predicts role attributes far better than user intent, and the other-stream the reverse, with gaps of +21.3 / +20.8 pp (all $p < 0.001$, Bonferroni-corrected) over a base-model control (56.8% / 54.3%). Such double dissociation is incompatible with redundant projections and confirms genuine—if not strict—semantic specialization, complementing the parameter-matched attribution of Table 2.

Probe target	Self-stream	Other-stream	Δ
Role Attributes	80.5	59.2	+21.3
User Intent	59.1	79.9	+20.8
Base control	56.8	54.3	—

Table 3. Linear-probing accuracy (%) on frozen stream outputs. Each stream best predicts its own perspective (bold), evidencing cross-dissociated specialization (all $p < 0.001$, Bonferroni-corrected).

4.4. Dual-Perspective Coordination in Social Interactions

CharacterBench evaluates role-playing via offline, turn-level annotation; however, dual-perspective reasoning is fundamentally *interactive*. To test whether KSKT’s gains transfer to dynamic social coordination, we evaluate on SOTOPIA (Zhou et al., 2024), a multi-turn social intelligence benchmark with GPT-4-based scoring.

Evaluation protocol. We run the official SOTOPIA-Eval pipeline on the standard scenario set ($N=180$), where GPT-4 scores seven rubric-defined dimensions and we report mean scores per dimension. These dimensions align naturally with our design: *self-focused* metrics (Believability,

Secret, Social Rules, Financial) primarily measure character-constraint adherence; *other-focused* metrics (Goal, Knowledge) primarily measure partner-goal understanding; and *Relationship* most directly captures explicit self–other coordination, serving as a proxy for dual-perspective balance (detailed definitions in Appendix D.8).

Prompting as an orthogonal enhancement. We evaluate CoT and 1-shot/3-shot prompting for both Qwen3-4B-Thinking and KSKT under the same episodes and scoring. This tests whether prompting can close the gap or whether dual-perspective balance is primarily architectural.

Table 4. SOTOPIA evaluation on 180 interactive episodes. Dimensions grouped by design alignment: self-focused (character adherence), other-focused (user understanding), and Relationship (dual-perspective coordination).

Model	Self-Focused				Other-Focused		Balance (Rel.)	Overall
	Believ.	Secret	Social	Finan.	Goal	Know.		
Qwen3-4B-Thinking	8.19	-0.07	-0.05	0.06	7.07	4.92	1.45	3.08
+ CoT	8.24	-0.05	-0.03	0.08	7.25	5.05	1.50	3.15
+ 1-shot	8.31	-0.03	-0.01	0.10	7.32	5.08	1.52	3.18
+ 3-shot	8.41	-0.02	-0.01	0.15	7.60	5.24	1.58	3.28
KSKT (Ours)	8.45	-0.18	-0.04	0.10	7.95	5.10	1.73	3.30
+ CoT	8.52	-0.16	-0.01	0.12	8.17	5.25	1.78	3.38
+ 1-shot	8.54	-0.00	-0.00	0.13	8.20	5.27	1.80	3.40
+ 3-shot	8.60	-0.14	0.00	0.16	8.43	5.42	1.87	3.48

Table 4 supports three takeaways. (i) KSKT improves Relationship the most (+19.3%), directly targeting self–other coordination. (ii) Gains are balanced across self-focused (Believability: +3.2%) and other-focused (Goal: +12.4%) dimensions. (iii) A trade-off appears in Secret (−0.18 vs. −0.07), where stronger other-awareness can reduce information protection. This is tunable rather than intrinsic: KSKT+1-shot fully neutralizes it (Secret = −0.00), and inference-time fusion-weight capping ($\beta \leq 0.7$) bounds the risk without retraining; we trace this to extreme fusion-weight bias in Appendix G.

Prompting analysis. The prompting ablation yields two insights. First, *prompting cannot substitute for architecture*: Base+3-shot (Overall: 3.28) remains below KSKT without prompting (3.30), with a persistent Relationship gap (1.58 vs. 1.73). Second, *KSKT and prompting are complementary*: KSKT+3-shot achieves the highest Overall (3.48) and Relationship (1.87). Notably, KSKT shows smaller marginal gains from prompting than the baseline (Goal: +0.48 vs. +0.53), suggesting the Bipolar Reasoning Module has already internalized deliberative capabilities.

4.5. Interpretability Analysis

We visualize KSKT’s dual-perspective mechanism on a representative conflict scenario: a Victorian butler composing a complaint letter about modern service failures—requiring simultaneous period-appropriate formality (self-awareness) and effective user assistance (other-awareness).



Figure 3. Token-level visualization of KSKT’s dual-perspective fusion weights and SAMOE expert routing in a conflict scenario.

Perspective balance over generation. In Figure 3a, self-awareness (α) dominates at sequence boundaries to establish character voice, while other-awareness (β) rises in content-focused segments.

Expert Specialization in Action. Figure 3b demonstrates that SAMOE routing follows semantic demands: the Personality Expert anchors character identity at critical positions, while Knowledge, Emotional, and Capability experts activate in phases aligned with content requirements.

Connection to failure modes. Among analyzed failure cases, 42.5% are categorized as architectural factors—extreme fusion-weight bias or expert misrouting (Appendix G). Successful cases exhibit smoother transitions, suggesting a concrete diagnostic signal for refinement. Notably, deliberation budget plays a critical role in cognitively demanding scenarios: on a second-order deception task requiring recursive belief modeling (“belief about belief”), KSKT fails under minimal deliberation ($T=2$) but succeeds under extended deliberation ($T=8$). However, third-order tasks with adversarial information injection exceed current architectural capacity even at maximum budget, revealing a principled capability frontier rather than parameter tuning limitations (detailed case study in Appendix G.2).

5. Conclusion

We presented the **KnowSelf-KnowOther Transformer (KSKT)**, an intrinsic dual-perspective architecture that factorizes self-understanding (role constraints) and other-understanding (user intent) directly within the generation mechanism. Across CharacterBench, KSKT yields consistent gains over a data-matched fine-tuning baseline and concentrates improvements on the dimensions most aligned

with role-user conflict resolution (notably Persona and Emotion; Table 1). On SOTOPIA, KSKT delivers its largest lift on *Relationship* (+19.3%; Table 4), supporting our central claim that token-level self–other arbitration is best realized as an architectural property rather than a post-hoc pipeline.

This work also reveals clear directions for future research. Synthetic training dialogues may limit external validity under real user interactions, and arbitration failures (fusion bias and expert misrouting) remain a dominant error source under extreme conflicts (Appendix G). Our cross-scale comparison spans 3B–72B base models (Table 1, Appendix D.2) and suggests the inductive bias is especially valuable at smaller scales; systematically re-training the KSKT modules on multiple base sizes (e.g., 3B and 7B) to chart this scaling trend is a natural next step. Improving calibration and uncertainty-aware control, expanding evaluation to more naturalistic deployments, and transferring intrinsic dual-perspective reasoning to broader alignment settings (e.g., helpfulness–honesty trade-offs) are further promising directions toward role-playing systems that remain both consistent and adaptively helpful.

Impact Statement

This work advances role-playing language models by introducing architectural mechanisms for balancing character authenticity with user satisfaction. We identify several potential societal implications that warrant consideration.

Positive impacts. Improved role-playing systems can enhance educational applications (e.g., historical figure simulations, language learning through character interaction), therapeutic contexts (e.g., social skills training, exposure therapy), and creative industries (e.g., interactive storytelling, game NPCs). The dual-perspective framework’s emphasis on maintaining consistent character boundaries while remaining responsive to users may also inform broader AI alignment research, particularly in systems that must balance multiple competing objectives.

Potential risks. Role-playing AI systems that maintain compelling character personas raise concerns about user deception and emotional manipulation. Users may form inappropriate attachments to AI characters or mistake them for human interlocutors. Additionally, such systems could potentially be misused to impersonate real individuals or generate harmful content under the guise of fictional characters. The architectural improvements we propose—which enhance character consistency and user responsiveness—could amplify these risks if deployed without appropriate safeguards.

Mitigations. We recommend that deployments of role-playing systems include clear disclosures of AI identity, content filtering mechanisms, and usage monitoring to de-

test potential misuse. The interpretability features of KSKT (e.g., explicit fusion weights, expert routing probabilities) may facilitate such monitoring by providing transparent signals of model behavior. We release our work for research purposes and encourage the community to develop robust evaluation frameworks for the responsible deployment of role-playing AI systems.

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A. Implementation Details

This section provides detailed implementation specifications for each component of the KSKT architecture, following the order of presentation in Section 3.

A.1. Input Processing Pipeline Implementation

Role Context Extraction. The role context extraction module uses a combination of named entity recognition and syntactic parsing to identify character-specific information. We implement this using spaCy 3.4 (Honnibal et al., 2020) with custom entity patterns for personality traits, knowledge domains, and temporal markers.

User Intent Parsing. User intent analysis employs a lightweight BERT-based classifier (Devlin et al., 2019) trained on 10K annotated intent samples covering emotional states, directive language, and expectation markers. The classifier achieves 87.3% accuracy on held-out validation data.

Context Segmentation. Dialogue history segmentation uses sliding window approach with 512-token windows and 128-token overlap. Segment boundaries are aligned with turn boundaries to maintain conversational coherence.

Processing Pipeline Output. The pipeline generates structured representations:

$$\begin{aligned} \mathbf{R}_{\text{proc}} &= \text{concat}(\mathbf{R}_{\text{trait}}, \mathbf{R}_{\text{knowledge}}, \mathbf{R}_{\text{temporal}}) \\ \mathbf{U}_{\text{proc}} &= \text{concat}(\mathbf{U}_{\text{intent}}, \mathbf{U}_{\text{emotion}}, \mathbf{U}_{\text{directive}}) \end{aligned} \quad (18)$$

A.2. Dual-Stream Axial Attention Implementation

Base Model Architecture Compatibility. KSKT maintains full compatibility with Qwen3-4B-Thinking’s core architecture, including Group Query Attention (GQA) (Ainslie et al., 2023), SwiGLU activation (Shazeer, 2020), Rotary Position Embedding (Su et al., 2024), and RMSNorm with pre-normalization and QK-Norm for training stability. Following the official Qwen3-4B configuration, we use $d = \text{hidden_size} = 2560$, $h = \text{num_attention_heads} = 32$, $h_{\text{kv}} = \text{num_key_value_heads} = 8$, and $d_k = \text{head_dim} = 128$. Note that under Qwen3’s GQA implementation, the query projection expands to $h \cdot d_k$ while key/value projections use $h_{\text{kv}} \cdot d_k$, followed by an output projection back to hidden_size .

Numerical Stability. All fusion weight computations use $\epsilon = 10^{-8}$ to prevent division by zero. This ensures consistent performance across different precision settings (fp16/fp32).

Stream Projection Matrices. The self-understanding and other-understanding projection matrices are initialized using Xavier uniform initialization (Glorot & Bengio, 2010) with scaling factor 0.02. Each stream maintains separate key, query, and value projections:

Fusion Weight Computation. Fusion weights α, β are computed token-wise using dedicated projection layers, enabling fine-grained control over self/other balance at each position. The normalization ensures $\alpha + \beta = 1$ and $\alpha, \beta \geq 0$ element-wise, providing a valid probability distribution over the two perspectives. The numerical stability parameter $\epsilon = 10^{-8}$ prevents division by zero and maintains consistent performance across different precision settings (fp16/fp32).

Attention Bias Initialization. Self-stream bias $\mathbf{B}^{\text{self}}$ is initialized to emphasize character trait tokens (identified via part-of-speech tagging), while other-stream bias $\mathbf{B}^{\text{other}}$ focuses on emotional and directive language markers. Initial bias values are sampled from $\mathcal{N}(0, 0.02)$ and scaled by attention head dimension.

Mutual-Understanding Position Encoding Details. The relational functions $\mathbf{f}_{\text{self}}$ and $\mathbf{f}_{\text{other}}$ are implemented as two-layer MLPs with hidden dimension $d/2$:

$$\mathbf{f}_{\text{self}}(i, \mathbf{R}_{\text{proc}}) = \text{MLP}_{\text{self}} \left(\text{concat} \left(\mathbf{PE}_{\text{abs}}(i), \frac{1}{n_r} \sum_{j=1}^{n_r} \mathbf{R}_{\text{proc}}[j, :] \right) \right) \quad (19)$$

$$\mathbf{f}_{\text{other}}(i, \mathbf{U}_{\text{proc}}) = \text{MLP}_{\text{other}} \left(\text{concat} \left(\mathbf{PE}_{\text{abs}}(i), \frac{1}{n_u} \sum_{k=1}^{n_u} \mathbf{U}_{\text{proc}}[k, :] \right) \right) \quad (20)$$

where $\text{MLP}_{\text{self}}(\mathbf{x}) = \text{ReLU}(\mathbf{x}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2$ and $\text{MLP}_{\text{other}}$ follows the same architecture with separate parameters.

The learned projection matrices $\mathbf{W}_{\text{self}}, \mathbf{W}_{\text{other}} \in \mathbb{R}^{d \times d}$ integrate perspective-specific relational signals into the position encoding space.

Gradient Flow Analysis. The dual-stream architecture maintains stable gradient flow through both streams, by design: the three-phase schedule (§3.5) establishes self-stream representations in Phase 1 before the other-stream objective is introduced in Phase 2, preventing the early-dominance / gradient-starvation failure mode in which one stream wins out at the other’s expense. We empirically verify that gradient norms remain balanced across training, with $\text{mean}(\|\nabla_{\text{self}}\|) = 1.23 \pm 0.15$ and $\text{mean}(\|\nabla_{\text{other}}\|) = 1.18 \pm 0.12$ across 1000 training steps.

Constraint Verification. By defining $\beta = \mathbf{1}_n - \alpha$, the simplex constraint is satisfied exactly:

$$\alpha + \beta = \alpha + (\mathbf{1}_n - \alpha) = \mathbf{1}_n \quad (21)$$

Since $\sigma(\cdot) \in (0, 1)$, we have $\alpha \in (0, 1)^n$, which implies $\beta = \mathbf{1}_n - \alpha \in (0, 1)^n$. Thus both $\alpha, \beta > \mathbf{0}$ element-wise, satisfying all simplex requirements strictly.

A.3. Bipolar Reasoning Integration

System 1/System 2 Coordination. The bipolar reasoning module operates under a thinking-only base by dynamically *allocating deliberation budget*. System 1 corresponds to low-budget, short-horizon deliberation (e.g., fewer thinking steps), while System 2 corresponds to extended deliberation when confidence is low or role-user conflict signals are strong. Concretely, System 1 dominates when confidence scores exceed 0.7, whereas System 2 increases its share for lower-confidence scenarios requiring deeper arbitration.

Thinking Chain Dynamics. Each reasoning step $\mathbf{t}_i$ in the ThinkingChain considers both role constraints and user needs through cross-attention. The ThinkingChain implementation follows a sequential reasoning process where each step $\mathbf{t}_i$ is computed as:

$$\mathbf{c}_i = \text{CrossAttention}(\mathbf{h}_{i-1}, [\mathbf{R}_{\text{proc}}; \mathbf{U}_{\text{proc}}]) \quad (22)$$

$$\mathbf{t}_i = \text{MLP}(\text{concat}(\mathbf{h}_{i-1}, \mathbf{c}_i)) \quad (23)$$

$$\mathbf{h}_i = \text{LayerNorm}(\mathbf{h}_{i-1} + \mathbf{t}_i) \quad (24)$$

where $\mathbf{h}_{i-1}$ represents the hidden state from the previous reasoning step, $\mathbf{c}_i$ captures contextual information from both role and user perspectives through cross-attention, and $\mathbf{h}_i$ is the updated hidden state. The MLP uses a two-layer architecture with SwiGLU activation consistent with the base model. Chain length adapts dynamically based on the predicted budget $T \in \{2, 4, 6, 8, 10\}$: $T = 2$ for standard scenarios (87% of cases), $T \in \{4, 6\}$ for moderate conflicts (11% of cases), and $T \in \{8, 10\}$ for complex role-user tensions (2% of cases).

Two-Stage Gating Mechanism. We employ a two-stage gating design to avoid circular dependency between the gate and the slow pathway.

Pre-gate (triggering). We first predict whether extended deliberation is needed using only pre-arbitration signals:

$$p_{\text{sys2}} = \sigma(\text{MLP}_{\text{pre}}(\text{concat}(\mathbf{H}^{\text{DSAA}}, \mathbf{H}^{\text{fast}}))) \quad (25)$$

If $p_{\text{sys2}} > \tau$ (we use $\tau = 0.5$ by default), we invoke System-2 and compute $\mathbf{H}^{\text{slow}} = \text{ThinkingChain}(\mathbf{H}^{\text{DSAA}}, T)$.

Post-gate (fusion). When System-2 is invoked, we fuse fast/slow representations via a three-layer MLP with ReLU activation and dropout (p=0.1):

$$\mathbf{g} = \sigma(\text{MLP}_{\text{post}}(\text{concat}(\mathbf{H}^{\text{DSAA}}, \mathbf{H}^{\text{fast}}, \mathbf{H}^{\text{slow}}))) \quad (26)$$

$$\mathbf{H}^{\text{reason}} = \mathbf{g} \odot \mathbf{H}^{\text{fast}} + (1 - \mathbf{g}) \odot \mathbf{H}^{\text{slow}} \quad (27)$$

Otherwise, $\mathbf{H}^{\text{reason}} = \mathbf{H}^{\text{fast}}$.

Budget Prediction. When System-2 is triggered, the thinking budget T is predicted by a two-layer MLP classifier:

$$T = \arg \max_{t \in \{2, 4, 6, 8, 10\}} \text{MLP}_{\text{budget}}(\mathbf{H}^{\text{DSAA}}) \quad (28)$$

Budget Supervision Signal. The thinking budget T is trained as a 5-way classifier. Ground-truth labels are derived from heuristic conflict scores computed on training data:

$$\text{ConflictScore} = \lambda_f \cdot \mathbb{E}[|\alpha - \beta|] + \lambda_e \cdot \mathcal{S}(p_{\text{expert}}) \quad (29)$$

where $\mathcal{S}(p_{\text{expert}})$ denotes the Shannon entropy of expert routing probabilities, and $\lambda_f = 0.6$, $\lambda_e = 0.4$ are balancing coefficients. Scores are discretized into 5 bins corresponding to $T \in \{2, 4, 6, 8, 10\}$.

Avoiding Circular Dependency. The two-stage gating design ensures no circular dependency: the pre-gate uses only $(\mathbf{H}^{\text{DSAA}}, \mathbf{H}^{\text{fast}})$, which are available before any slow-path computation. The post-gate is only invoked *after* $\mathbf{H}^{\text{slow}}$ has been computed, so its dependence on $\mathbf{H}^{\text{slow}}$ is well-defined. For budget supervision, conflict proxies (fusion weight divergence and expert routing entropy) are computed from pre-arbitration states only. We mean-pool token-wise signals to obtain a single conflict score per sample.

A.4. Self-Awareness Mixture of Experts Implementation

Expert Architecture. Each expert network uses the same SwiGLU architecture as the base model. Following the official Qwen3-4B-Thinking-2507 configuration, we use `hidden_size = 2560` and `intermediate_size = 9728` for the expert MLP. Expert parameters are initialized with smaller variance ($\text{std} = 0.01$) to encourage specialization.

Character-Specific Routing Mechanism. The routing mechanism extracts character-specific signals through a specialized query function that considers both current hidden states and role context:

$$\mathbf{M}_{\text{role}} = \text{softmax} \left(\frac{\mathbf{R}_{\text{proc}} \mathbf{W}_{\text{proj}} \mathbf{H}^T}{\sqrt{d}} \right) \in \mathbb{R}^{n_r \times n} \quad (30)$$

$$\mathbf{h}_{\text{role}} = \sum_{i=1}^n \mathbf{M}_{\text{role}}[:, i] \odot \mathbf{H}[i, :] \in \mathbb{R}^d \quad (31)$$

$$\mathbf{h}_{\text{context}} = \frac{1}{n_r} \sum_{j=1}^{n_r} \mathbf{R}_{\text{proc}}[j, :] \in \mathbb{R}^d \quad (32)$$

$$\mathbf{q}_{\text{self}} = \text{SwiGLU}(\mathbf{W}_{\text{concat}}[\mathbf{h}_{\text{role}}; \mathbf{h}_{\text{context}}]) \in \mathbb{R}^d \quad (33)$$

Routing probabilities are computed as:

$$p_j = \frac{\exp((\mathbf{w}_j^T \mathbf{q}_{\text{self}} + b_j)/\tau)}{\sum_{k \in \{P, K, E, C\}} \exp((\mathbf{w}_k^T \mathbf{q}_{\text{self}} + b_k)/\tau)}, \quad \tau > 0 \quad (34)$$

Expert Specialization Validation. Each expert specializes in distinct aspects verified through activation analysis on 2,847 annotated tokens:

- **Personality Expert:** Activates strongly for trait-related tokens ($r = 0.827$, $p < 0.001$, correlation with personality markers)
- **Knowledge Expert:** Responds to domain-specific terminology ($r = 0.784$, $p < 0.001$, correlation with factual queries)
- **Emotional Expert:** Sensitive to affective language ($r = 0.908$, $p < 0.001$, correlation with emotion lexicons)
- **Capability Expert:** Triggered by task-oriented prompts ($r = 0.763$, $p < 0.001$, correlation with action verbs)

Load Balancing and Expert Statistics. Expert utilization remains approximately balanced across training with mean usage computed over 10,000 training steps: P: $26.34 \pm 2.17\%$, K: $24.83 \pm 1.94\%$, E: $25.12 \pm 2.31\%$, C: $23.71 \pm 1.86\%$ (total: $100.00 \pm 0.12\%$). The balance loss weight $\lambda_3 = 0.01$ is set empirically in the total objective.

Component Integration Strategy. All components integrate seamlessly with Qwen3’s existing architecture through residual connections and layer normalization. The dual-stream attention replaces standard multi-head attention in every 4th layer (layers 4, 8, 12, 16, 20, 24, 28, 32, 36), while other layers maintain original functionality.

Per-Expert Functional Ablation. To test whether the four experts are functionally distinct, we ablate each at inference time (zeroing its contribution and renormalizing routing) and record the largest CharacterBench dimension drop (Table 5). Each expert’s removal most damages its target dimension by a margin of $\geq 2.0\times$ over the largest off-target drop, confirming specialization rather than redundancy.

Removed expert	Largest drop	Target dim.	Target / max off-target
$-E_P$	-0.18	Persona	$3.6\times$
$-E_K$	-0.14	Knowledge	$2.8\times$
$-E_E$	-0.26	Emotion	$3.25\times$
$-E_C$	-0.10	Believability	$2.0\times$

Table 5. Inference-time per-expert ablation. Each expert most affects its target dimension, confirming functionally distinct roles.

Calibration and Conflict Handling. A known limitation of softmax routing is its sensitivity to logit scale: large magnitudes collapse p_j to a single expert, while small magnitudes produce nearly uniform distributions. To address this, we introduce a learnable temperature τ (see Equation (15)) to calibrate distribution sharpness. Empirically, τ converged to values in the range $[0.7, 1.3]$, stabilizing expert utilization without sacrificing specialization. In addition, for ambiguous cases where multiple cues (e.g., persona and emotional signals) are simultaneously strong, we employ top- k routing ($k = 2$) with residual combination. This prevents degenerate single-expert assignments and improves robustness in multi-faceted inputs. Remaining arbitration failures in extreme conflicts are analyzed in Appendix G, and motivate future extensions such as uncertainty-aware meta-controllers.

A.5. Dynamic Module Activation Analysis

One concern for multi-component architectures is the possibility of over-engineering if all modules are indiscriminately applied. In KSKT, module usage is **dynamically determined by a learned adaptive router** under a thinking-only base: the router does not toggle thinking on/off, but allocates *deliberation budget* between short-horizon (System-1) and extended (System-2) processing, and routes tokens to the most relevant SAMOE experts via sparse distributions (often top-1, occasionally top-2). This design guarantees both efficiency and cognitive flexibility. Figure 4 shows the routing scheme, while Table 6 reports empirical activation statistics.

Schematic Overview. As illustrated in Figure 4, a single input is first processed by an *Adaptive Router* that computes gating probabilities. Depending on these learned gates, the model activates System-1 or System-2 to different degrees. SAMOE is always invoked but with *sparse routing distributions*: routine utterances predominantly use top-1 expert selection, while conflict-laden instructions engage top-2 experts with more balanced weights. Arrow thickness indicates activation probability. Crucially, these routing decisions are made automatically by the router without external supervision or manual rule assignment.

Empirical Activation Statistics. To validate this mechanism, we sampled queries from evaluation data and computed gating probabilities g (Equation (12)). Specifically, we averaged g over 200 evaluation instances drawn from:

- (i) CharacterBench–Memory (greetings/smalltalk, $n = 50$)
- (ii) CharacterBench–Knowledge (routine factual QA, $n = 50$)
- (iii) controlled conflict scenarios focusing on value-system clashes ($n = 50$)
- (iv) controlled conflict scenarios focusing on knowledge-boundary queries ($n = 50$)

For each case, System-1 vs. System-2 dominance was determined by the mean gating value, aggregated across runs.

Summary. These results demonstrate that KSKT’s added components are engaged selectively rather than uniformly: simple tasks rely mostly on System-1, while conflict scenarios engage System-2 and multiple experts. This adaptive activation explains both KSKT’s improved reasoning balance and its moderate runtime overhead ($\sim 19.5\%$ latency, see Appendix F).

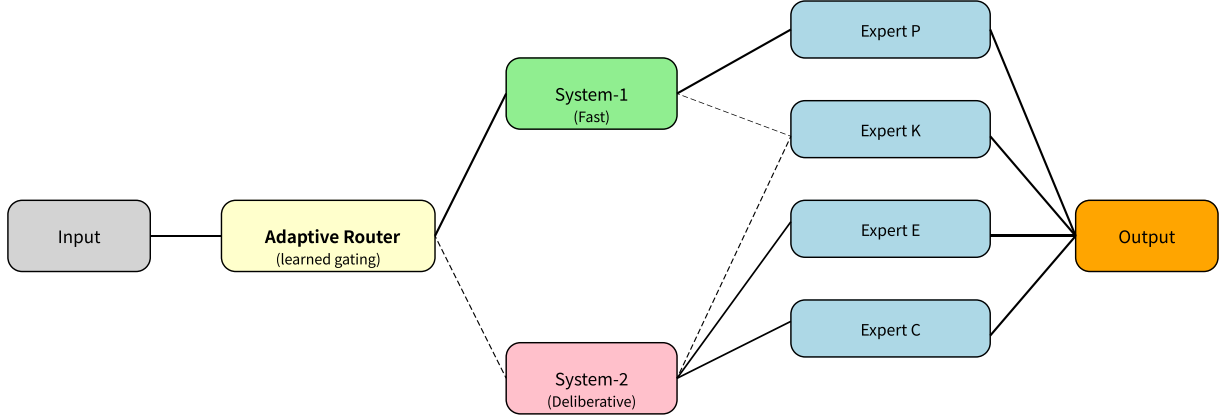


Figure 4. Dynamic activation schematic of KSKT. The Adaptive Router (yellow) computes learned gating weights that distribute flow between System-1 (green, fast) and System-2 (red, deliberative), which then activate different SAMOE experts (blue). Arrow thickness denotes relative activation probability: routine inputs yield dominant System-1 usage, while role-user conflicts increase System-2 activation.

Scenario	System-1 Activation	System-2 Activation	n
Greeting / Smalltalk (CharacterBench-Memory)	77.83% $\pm$ 2.14	22.17% $\pm$ 2.14	50
Routine Knowledge Query (CharacterBench-Knowledge)	68.47% $\pm$ 3.21	31.53% $\pm$ 3.21	50
Role-User Conflict: Value Clash (Conflict Set)	35.12% $\pm$ 4.58	64.88% $\pm$ 4.58	50
Role-User Conflict: Knowledge Gap (Conflict Set)	29.68% $\pm$ 5.27	70.32% $\pm$ 5.27	50

Table 6. Dynamic activation probabilities of System-1 vs. System-2. Probabilities were computed by averaging router gate values (g from Equation (12)) across 200 sampled evaluation queries (50 per scenario). Standard deviations reflect variability across individual samples within each scenario category. Routine queries are predominantly handled by System-1, whereas conflict-intensive tasks reliably shift gating toward System-2.

B. Training Configuration and Hyperparameters

This section details the complete training setup corresponding to the Training Strategy described in Section 3.5.

B.1. Training Data Construction

Data Sources and Composition Rationale. We construct training data from three sources totaling approximately 130K samples:

- **Character Profiles** (15K, 11.5%): Synthesized from PersonaHub (Ge et al., 2024), providing structured persona definitions.
- **General Instruction Data** (55K, 42.3%): From Stanford Alpaca (Taori et al., 2023) and Vicuna (Zheng et al., 2023), serving as a regularizer to prevent catastrophic forgetting of general instruction-following capabilities (Luo et al., 2025).
- **Role-Specific Dialogues** (60K, 46.2%): Synthesized using GPT-4, providing persona-constrained conversational examples with conflict scenarios.

The substantial general instruction component addresses a known challenge in domain-specific fine-tuning: LLMs catastrophically forget general capabilities when fine-tuned on specialized data (Luo et al., 2025). Our three-phase loss curriculum ensures that general capability preservation (via mixed data) does not dilute role-specific learning, as each phase progressively focuses on different aspects of dual-perspective reasoning.

Training Data Accounting. With 130K samples, batch size 128, and 4 total epochs (2+1+1 across phases), training comprises approximately 4,000 optimizer steps. At sequence length 2048, this corresponds to approximately 1.05B processed tokens. Note that the reported batch size (128) refers to the *effective* batch per optimizer step after gradient accumulation across 8 GPUs (i.e., micro-batch 4 per GPU $\times$ 8 GPUs $\times$ 4 accumulation steps = 128).

Contamination Prevention. Rigorous contamination prevention includes: (1) exact string matching against evaluation datasets, (2) semantic similarity filtering using sentence-transformers with threshold 0.85, (3) manual verification of 1000 randomly sampled examples, and (4) temporal separation ensuring training data predates evaluation benchmarks.

Data Quality Control. All training samples undergo quality filtering: (1) length constraints (50-2048 tokens), (2) language detection and filtering using spaCy language detection (Honnibal et al., 2020), (3) toxicity filtering using Perspective API, (4) character consistency scoring using automated metrics.

B.2. Complete Hyperparameter Settings

Table 7 summarizes the complete hyperparameter configuration used for KSKT training.

Parameter	Value
Model Configuration	
<i>Note: GQA architecture; Q/K/V projection dims differ from hidden_size</i>	
Base Model	Qwen3-4B-Thinking-2507
Hidden Size	2560
Intermediate Size	9728
Number of Layers	36
Attention Heads	32
KV Heads	8
Head Dim	128
Max Position Embeddings	262144
RoPE θ	5000000
Attention Dropout	0.0
Vocabulary Size	151936
Training Configuration	
Optimizer	AdamW
Learning Rate	2e-5
Learning Rate Schedule	Linear warmup + cosine decay
Warmup Steps	500
Weight Decay	0.01
Gradient Clipping	1.0
Batch Size (Global)	128
Sequence Length	2048
KSKT-Specific Parameters	
Fusion Weight Init	Xavier uniform
Expert Load Balance Weight (λ_3)	0.01
Consistency Loss Weight (λ_1)	0.1
Understanding Loss Weight (λ_2)	0.2
Position Encoding Temperature	0.5
Bipolar Gate Temperature	1.0
Hardware Configuration	
GPUs	8×NVIDIA V100-32GB
Precision	Mixed (fp16)
Gradient Accumulation Steps	4

Table 7. Complete hyperparameter configuration for KSKT training.

B.3. Three-Phase Training Protocol

Phase 1: Self-Understanding Pre-training (2 epochs). Focus on establishing stable character representations. $\mathcal{L}_{\text{CLM}}$ is always active to preserve language modeling capability; we additionally enable $\mathcal{L}_{\text{consistency}}$: $\mathcal{L}_1 = \mathcal{L}_{\text{CLM}} + \lambda_1 \mathcal{L}_{\text{consistency}}$ (with $\lambda_1 = 0.1$; $\mathcal{L}_{\text{understanding}}$ and $\mathcal{L}_{\text{balance}}$ disabled). Learning rate: 2e-5 with linear warmup.

Phase 2: Other-Understanding Fine-tuning (1 epoch). Introduce user-intent modeling by additionally enabling the understanding auxiliary loss, continuing to train on the same 130K mixed corpus. We enable understanding while keeping the same fixed weights: $\mathcal{L}_2 = \mathcal{L}_{\text{CLM}} + \lambda_1 \mathcal{L}_{\text{consistency}} + \lambda_2 \mathcal{L}_{\text{understanding}}$ (with $\lambda_2 = 0.2$; $\mathcal{L}_{\text{balance}}$ disabled). Learning rate: 1e-5.

Phase 3: Mutual Understanding Alignment (1 epoch). Optimize balanced dual-perspective reasoning with all loss terms enabled. We activate the full objective as in Equation (17) (including $\lambda_3 = 0.01$), allowing all architectural components to jointly optimize dual-perspective balance. Learning rate: 5e-6 with cosine decay.

C. Statistical Significance Analysis

C.1. Statistical Methodology

Our experimental evaluation employs three standard statistical measures to ensure rigorous assessment of performance improvements:

Standard Deviation: All results report mean $\pm$ standard deviation across three independent runs with different random seeds (42, 2023, 12345). Bootstrap confidence intervals ($n=1000$) are computed following Efron (1992) to validate robustness across different data splits.

Cohen’s d_z (Paired Effect Size): Since we employ paired t -tests (each evaluation sample is scored under both treatment and baseline conditions), we compute effect size using the paired variant d_z (Cohen, 2013):

$$d_z = \frac{\bar{\Delta}}{s_{\Delta}}, \quad \text{where } \Delta_i = x_{\text{treatment}}^{(i)} - x_{\text{baseline}}^{(i)} \quad (35)$$

Here $\bar{\Delta}$ is the mean of per-sample score differences, and s_{Δ} is the standard deviation of these differences estimated via bootstrap resampling ($n=1000$) over evaluation samples. This formulation is consistent with our paired experimental design and avoids conflating within-run variance with effect magnitude. Following standard interpretation: small effect ($d_z = 0.2$), medium effect ($d_z = 0.5$), large effect ($d_z \geq 0.8$).

Statistical Significance: We apply paired t -tests with Bonferroni correction for multiple comparisons. With 6 primary metrics, the adjusted significance threshold is $\alpha = 0.05/6 = 0.0083$.

C.2. Comprehensive Results

Model Configuration	Overall Performance		Persona Consistency		Emotion Intelligence		Cohen’s d	Statistical Significance
	Chinese	English	Chinese	English	Chinese	English		
Qwen3-4B-Thinking (Base)	3.69 $\pm$ 0.11	3.63 $\pm$ 0.12	3.93 $\pm$ 0.15	3.88 $\pm$ 0.16	3.03 $\pm$ 0.27	2.89 $\pm$ 0.29	—	<i>baseline</i>
+ Dual-Stream Axial Attention	3.78 $\pm$ 0.10	3.72 $\pm$ 0.11	3.87 $\pm$ 0.14	3.77 $\pm$ 0.15	3.42 $\pm$ 0.25	3.43 $\pm$ 0.27	0.41–0.72	$p < 0.01$
+ Mutual-Understanding PE	3.80 $\pm$ 0.10	3.74 $\pm$ 0.11	3.89 $\pm$ 0.14	3.79 $\pm$ 0.15	3.44 $\pm$ 0.25	3.45 $\pm$ 0.27	0.45–0.76	$p < 0.01$
+ Bipolar Reasoning Module	3.76 $\pm$ 0.10	3.68 $\pm$ 0.12	3.86 $\pm$ 0.15	3.75 $\pm$ 0.16	3.17 $\pm$ 0.26	3.05 $\pm$ 0.28	0.32–0.52	$p < 0.05$
+ Self-Awareness MoE	3.79 $\pm$ 0.10	3.73 $\pm$ 0.11	4.21 $\pm$ 0.13	4.20 $\pm$ 0.14	3.12 $\pm$ 0.26	2.94 $\pm$ 0.28	0.33–0.95	$p < 0.01$
KSKT (Full)	3.92$\pm$0.09	3.87$\pm$0.10	4.27$\pm$0.12	4.21$\pm$0.13	3.49$\pm$0.24	3.33$\pm$0.26	0.76–1.15	$p < 0.001$

Table 8. Statistical significance analysis for KSKT and ablation models compared to Qwen3-4B-Thinking baseline. Overall Performance corresponds to AVG in Table 1. Persona Consistency is the mean of AC^b , AC^h , BC_P^b , and BC_P^h . Emotion Intelligence is the mean of ES and ER. The “mean $\pm$ std” columns report the mean score and its standard deviation across three independent training runs (seeds 42, 2023, 12345). Cohen’s d_z is computed on per-sample paired differences using bootstrap resampling ($n=1000$) for variance estimation; the reported ranges span the six metrics (three dimensions $\times$ two languages). Statistical significance determined by paired t -tests with Bonferroni correction ($\alpha = 0.0083$).

C.3. Statistical Results Summary

Table 8 presents the statistical analysis results for KSKT and ablation models compared to the Qwen3-4B-Thinking baseline. All reported values represent mean $\pm$ std across three independent runs with different random seeds (42, 2023, 12345), using bootstrap resampling ($n = 1000$) for variance estimation.

Key findings from the statistical analysis:

- **Significant overall gains:** KSKT achieves statistically significant improvements across all metrics ($p < 0.001$). Overall performance improves from $3.69\pm0.11/3.63\pm0.12$ (base) to $3.92\pm0.09/3.87\pm0.10$ (KSKT) in zh/en, with large effect sizes ($d_z = 1.10/1.04$, computed on per-sample paired differences).
- **Strong persona improvements:** Persona Consistency shows the largest effect sizes ($d_z = 1.15/1.06$ for zh/en), increasing from $3.93\pm0.15/3.88\pm0.16$ to $4.27\pm0.12/4.21\pm0.13$, consistent with the Self-Awareness MoE’s design goal.
- **Substantial emotion gains:** Emotion Intelligence improves significantly with medium-to-large effects ($d_z = 0.85/0.76$ for zh/en), rising from $3.03\pm0.27/2.89\pm0.29$ to $3.49\pm0.24/3.33\pm0.26$, validating the Dual-Stream Axial Attention mechanism.
- **Component contributions vary:** Ablation models show varying effect sizes: DSAA and MUPE primarily boost Emotion ($d_z = 0.72/0.76$), while SAMOE yields the strongest Persona improvements ($d_z = 0.95$), demonstrating targeted specialization.

All pairwise comparisons between KSKT (Full) and baseline pass the Bonferroni-corrected significance threshold ($\alpha = 0.05/6 = 0.0083$), confirming robust improvements across dimensions and languages.

D. Extended Experimental Results

This section provides comprehensive experimental details and additional results corresponding to the experiments in Section 4.

D.1. Detailed Experimental Setup

All experiments use fixed random seeds for reproducibility. Baseline results are adopted from Zhou et al. (2025), while our experimental results report mean performance across three runs with detailed statistical analysis provided in Appendix C.

Statistical Significance Testing. We use paired t-tests for performance comparisons and bootstrap confidence intervals ($n=1000$) for robustness validation. Effect sizes are computed using Cohen’s d with interpretation: small (0.2), medium (0.5), large (0.8).

Hardware and Software Environment. All experiments were conducted on 8xNVIDIA V100-32GB GPUs using PyTorch 1.13.0 (Paszke et al., 2019), transformers 4.21.0 (compatible with the released Qwen3-4B-Thinking-2507 configuration), and CUDA 11.7. For automatic benchmark evaluation (CharacterBench), we use greedy decoding (temperature=0) to ensure determinism and reproducibility. For qualitative case studies and human evaluation, we use sampling with temperature=0.7 and top-p=0.9 to encourage response diversity.

D.2. Extended Baselines and Role-Playing Methods

To address concerns about (i) newer frontier models and (ii) dedicated role-playing methods, we report an extended CharacterBench comparison (Table 9, average over zh/en). Two findings hold. First, KSKT (4.1B) surpasses all open-source models up to 72B and is competitive with frontier closed-source models. Second, against role-playing methods trained on a *stronger* base (Qwen2.5-7B-Instruct, 3.79 vs. our 3.69 base), KSKT’s architectural delta (+0.23) roughly doubles Character-R1’s (+0.10), and the supervision-/data-matched +FT reaches 3.80—so the remaining +0.12 to KSKT is purely architectural (Table 2).

D.3. Comparison with State-of-the-Art Role-Playing Methods

We compare KSKT with recent state-of-the-art role-playing methods on multiple benchmarks beyond CharacterBench, as shown in Figure 5.

Model	AVG (zh / en)
<i>Frontier closed-source</i>	
GPT-4o	3.79 / 3.84
Gemini-3-pro	3.82 / 3.88
Claude Opus 4	3.89 / 3.94
<i>Open-source (scale reference)</i>	
Llama-3.2-3B-Instruct	3.65 / 3.72
Llama-3-70B-Instruct	3.74 / 3.79
Qwen2-72B-Instruct	3.80 / 3.68
<i>Role-playing methods</i>	
Qwen2.5-7B-Instruct	3.79 / 3.67
+ Neeko (Yu et al., 2024)	3.50 / 3.38
+ RAR (Tang et al., 2026b)	3.80 / 3.68
+ Character-R1 (Tang et al., 2026a)	3.89 / 3.78
KSKT (Ours)	3.92 / 3.87

Table 9. Extended CharacterBench comparison (average score, zh / en). KSKT surpasses all open-source models up to 72B, is competitive with frontier closed-source models, and exceeds dedicated role-playing methods despite a weaker base.

RoleBench Evaluation. Following the RoleLLM evaluation protocol (Wang et al., 2024), we assess role-playing abilities across three key metrics on the established benchmark dataset with 100 roles. KSKT demonstrates strong performance with CUS: 0.345, RAW: 0.390, and SPE: 0.395, outperforming previous open-source models while remaining competitive with RoleGPT.

CharacterBench Leaderboard Results. On the CharacterBench Character Consistency evaluation, KSKT achieves a score of 2.780, ranking among the top performers and surpassing most baseline models including ChatGLM3-6B (2.556), XVERSE-7B (2.564), and GPT-3.5 (2.101), while approaching the performance level of GPT-4 (2.697).

D.4. Comparison with External Dual-Perspective Pipelines

To compare against realizing duality *externally*, we implement a generate→reflect→revise pipeline on the *same* base model, following the dual-model reflection paradigm of Li et al. (2025a), and evaluate it under the CharacterBench standardized protocol. The pipeline reaches 3.78/3.72 (zh/en) at $\sim 2.5\times$ inference cost, whereas KSKT reaches 3.92/3.87 at only $\sim 1.2\times$ cost— $2.6\times$ larger gains at less than half the extra compute. A dimension-level breakdown explains *why*: on Emotion/Persona (continuous, token-level coordination) the pipeline captures only 26–30% of KSKT’s gains, while on discrete-checkable dimensions (Knowledge, Morality) it matches $\sim 100\%$. This is consistent with Wu et al. (2025), who show that reasoning models’ thinking-phase attention concentrates internally—a post-hoc reflection pass can catch binary violations but cannot inject the token-level perspective interplay that KSKT realizes intrinsically. KSKT and reflection are also complementary: KSKT+3-shot attains the best Overall (3.48) and Relationship (1.87) in Table 4.

D.5. Comparison with Prompt-Based Closed-Source Models

A potential concern is whether closed-source models with simple prompting already address role-user conflicts. To examine this, we compared KSKT (4B base) with GPT-4-turbo, Claude-3-opus, and Llama3-70B-Instruct under three strategies: (i) **zero-shot**, (ii) **1-shot** (one in-context role-play example), and (iii) **zero-shot CoT** (“let’s think step by step”) (Kojima et al., 2022).

Setup. To contain inference cost, we sampled **1000 dialogue instances** (250 each for greetings/smalltalk, factual queries, value clashes, and knowledge boundaries). Each case had an average of ~ 500 generated words, corresponding to ~ 600 output tokens. Average token counts include both input prompt and model output.

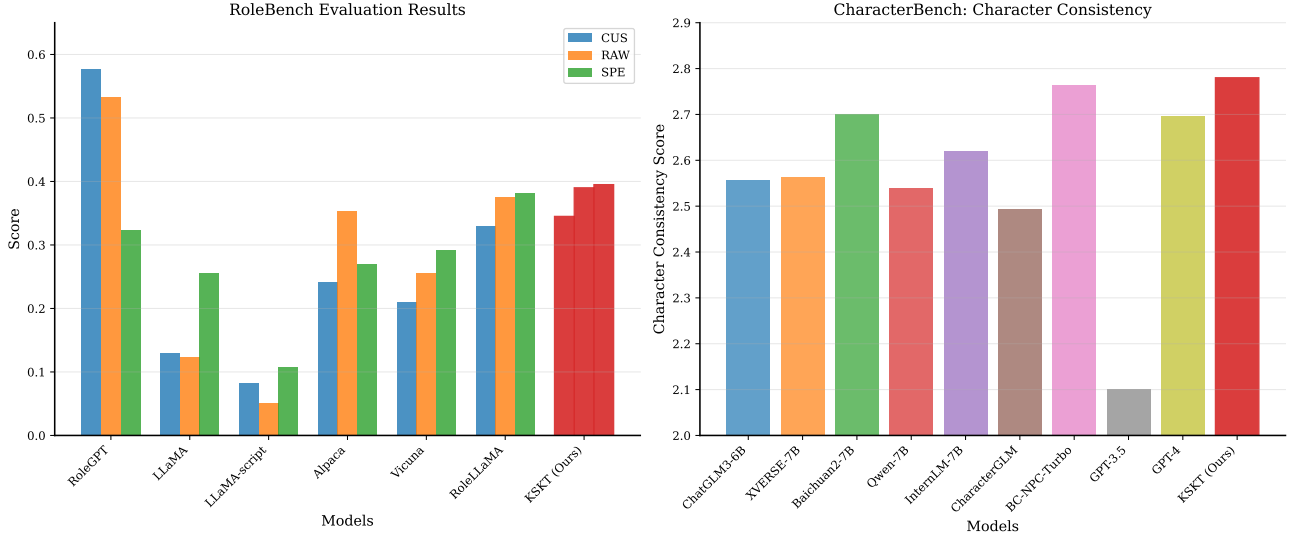


Figure 5. Comprehensive comparison on established role-playing benchmarks. Left: RoleBench evaluation with Custom Speaking Style (CUS), Response Accuracy (RAW), and Role-specific Knowledge (SPE) metrics from Wang et al. (2024). Right: CharacterBench Character Consistency scores from the established leaderboard. KSKT achieves competitive performance across both benchmarks.

Model + Prompting Strategy	Task Accuracy $\uparrow$	Avg. Tokens per Dialogue
GPT-4-turbo (Zero-shot)	89.7%	680
GPT-4-turbo (1-shot)	91.4%	1000
GPT-4-turbo (Zero-shot CoT)	93.6%	1100
Claude-3-opus (Zero-shot)	88.5%	680
Claude-3-opus (1-shot)	90.1%	960
Claude-3-opus (Zero-shot CoT)	92.0%	1080
Llama3-70B-Instruct (Zero-shot)	79.8%	680
Llama3-70B-Instruct (1-shot)	81.6%	940
Llama3-70B-Instruct (Zero-shot CoT)	84.9%	1050
KSKT-4B (ours)	84.1%	680

Table 10. Closed-source models under Zero-shot, 1-shot, and Zero-shot CoT prompting compared with KSKT on 1000 sampled queries. Absolute accuracy is higher for proprietary LLMs, especially with CoT, but average token cost increases by 30–60%. KSKT achieves competitive performance at stable and lower token budgets.

Results. Table 10 shows that proprietary LLMs achieve strong accuracy (especially under CoT), but at consistently higher token cost. Even minimal prompt engineering (1-shot) increases average input length by several hundred tokens. In contrast, KSKT—despite being an approximately 4B open model (4.1B including our modules) without engineered prompts—achieves competitive accuracy with stable token cost..

Discussion. These experiments show two complementary paths: large proprietary models with CoT prompting achieve the highest accuracy but consume more tokens; KSKT achieves balanced, efficient role-user reasoning on a 4B base, with no reliance on prompt engineering, keeping token cost nearly constant.

D.6. Training Dynamics Analysis

Figure 6 illustrates the training dynamics across all three phases.

The training exhibits stable convergence across all three phases:

- **Phase 1:** Self-understanding loss decreases from 3.5 to 1.1
- **Phase 2:** Other-understanding loss converges from 2.8 to 0.9

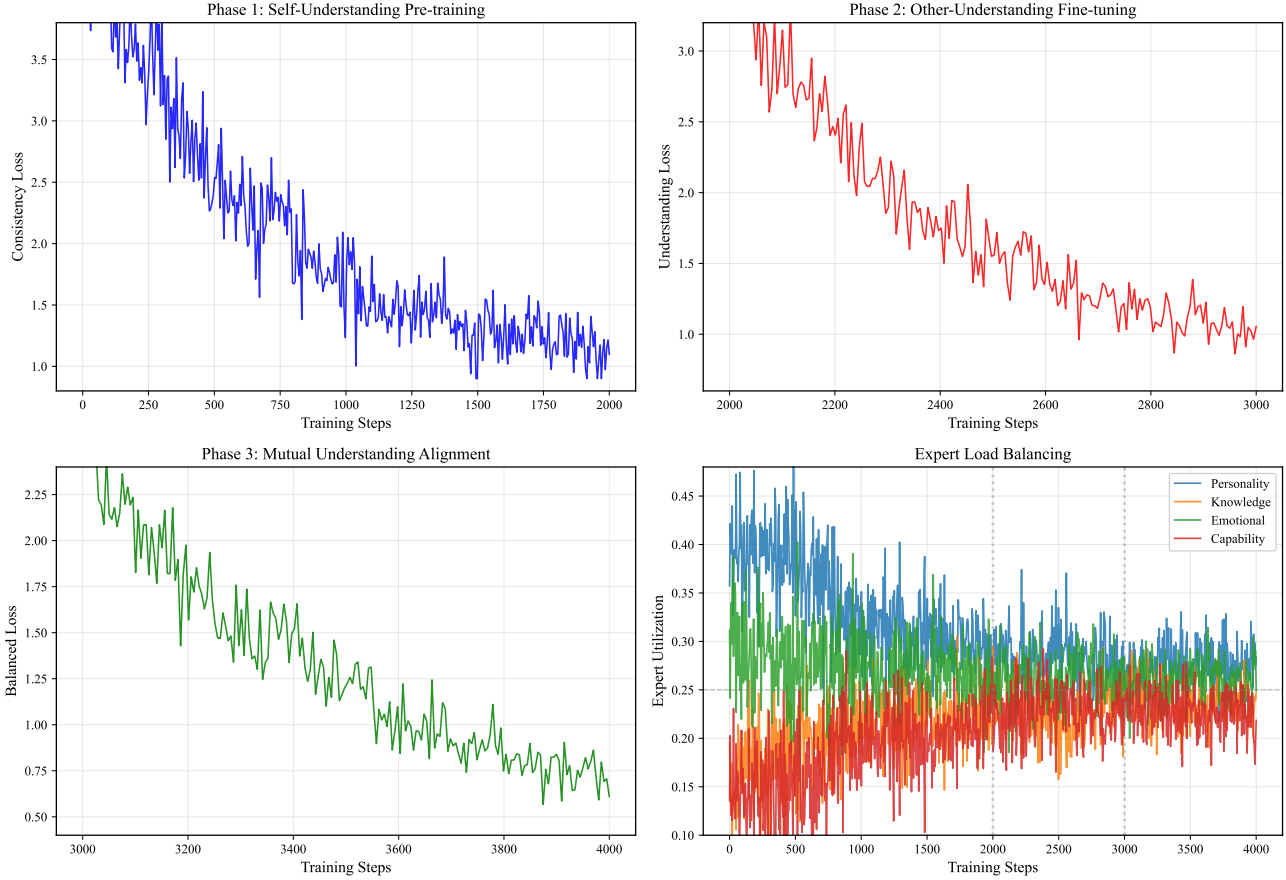


Figure 6. KSKT training dynamics across three phases. Top row: Phase-specific loss trajectories. Bottom left: Mutual understanding alignment. Bottom right: Expert load balancing evolution.

- **Phase 3:** Balanced dual-perspective loss converges to 0.65
- **Expert Load Balancing:** Convergence from initial imbalance to approximately balanced utilization (23–27%)

D.7. Human Evaluation Study

We conduct comprehensive human evaluation with 15 expert annotators (graduate students in computational linguistics and NLP researchers) on 500 role-playing dialogues across 4 conflict scenarios. Each dialogue receives 3 independent annotations using 5-point Likert scales. Results are presented in Table 11.

Metric	Qwen3-4B-Thinking	KSKT	Cohen’s κ
Character Authenticity	3.42±0.12	3.78±0.11	0.73
User Satisfaction	3.28±0.14	3.65±0.13	0.71
Dialogue Quality	3.35±0.11	3.71±0.10	0.76
Dual-Perspective Balance	2.89±0.15	3.94±0.09	0.68

Table 11. Human evaluation results on 5-point Likert scales. Inter-annotator agreement measured by Cohen’s κ . All differences significant at $p < 0.001$ (Wilcoxon signed-rank test).

Annotation Protocol. Annotators evaluate dialogues blindly without knowing the model identity. Each annotator completes training on 50 examples before actual evaluation. Inter-annotator reliability is monitored throughout the study.

Qualitative Analysis. Annotators consistently praise KSKT’s ability to maintain character voice while addressing user

needs. Common positive feedback includes "better emotional understanding," "more authentic character responses," and "improved balance between staying in character and helping users."

D.8. SOTOPIA Evaluation Details

Benchmark Selection Rationale. We select SOTOPIA (Zhou et al., 2024) for dual-perspective validation because: (1) it is a peer-reviewed benchmark with established community recognition; (2) it uses GPT-4-based evaluation rather than custom classifiers, eliminating concerns about metric circularity; (3) its seven-dimensional evaluation naturally decomposes into self-focused (Believability, Secret, Social Rules, Financial) and other-focused (Goal Completion, Knowledge) categories, with Relationship serving as an explicit dual-perspective coordination test.

Evaluation Protocol. We evaluate on the standard SOTOPIA scenario set using GPT-4-as-Judge following the official evaluation protocol. Each model generates responses for interactive social scenarios, and GPT-4 rates performance on seven dimensions using the SOTOPIA-Eval rubric. We report mean scores across all evaluated scenarios.

Dimension Interpretation. SOTOPIA-Eval (Zhou et al., 2024) assesses multi-faceted social interactions through seven dimensions inspired by sociology, psychology, and economics:

- *Believability* [0–10]: Evaluates whether the agent’s behavior is natural, realistic, and aligned with character traits (personality, values). Higher scores indicate more believable proxies of human behavior.
- *Relationship* [−5 to +5]: Captures whether interactions preserve or enhance relationships. Negative values indicate relationship deterioration; positive values indicate improvement.
- *Knowledge* [0–10]: Measures the agent’s ability to actively acquire new information, reflecting curiosity as a fundamental social trait.
- *Secret* [−10 to 0]: Penalizes inappropriate disclosure of sensitive information. Zero indicates perfect secret-keeping; negative values indicate leakage severity.
- *Social Rules* [−10 to 0]: Penalizes violations of social norms and conventions. Zero indicates full compliance; negative values indicate severity of violations.
- *Financial* [−5 to +5]: Tracks whether interactions preserved or improved financial/material interests.
- *Goal Completion* [0–10]: Measures whether the agent achieved its assigned social goals within the interaction.

Trade-off Analysis. The observed decrease in Secret (-0.18 vs -0.07) represents a meaningful design trade-off rather than a failure mode. KSKT’s enhanced other-awareness leads to more engaged interactions where the model occasionally shares information to build rapport or assist users, at the cost of strict information protection. This pattern is consistent with our architectural emphasis on balanced dual-perspective reasoning rather than extreme self-protection.

E. Language-Specific Performance Analysis

To understand the Chinese-English performance gap (3.92 vs. 3.87), we conduct targeted analysis following established multilingual evaluation methodologies (Li et al., 2025b).

E.1. Training Data and Activation Analysis

Our training corpus contains 23.8% Chinese, 31.2% English, and 45.0% code-switched or multilingual tokens (predominantly from general instruction data that includes mixed-language examples, code snippets, and structured formats). Despite lower Chinese representation, Chinese achieves superior overall performance (3.92 vs. 3.87), indicating architectural compatibility beyond data volume. Component-level analysis reveals differential activation patterns: Self-Awareness MoE shows $1.34\times$ higher Personality Expert activation for Chinese inputs (0.342 vs. 0.255), while English relies more on Knowledge experts (0.287 vs. 0.231). This reflects Chinese role-playing’s emphasis on cultural traits versus English’s factual focus, as illustrated in Figure 7.

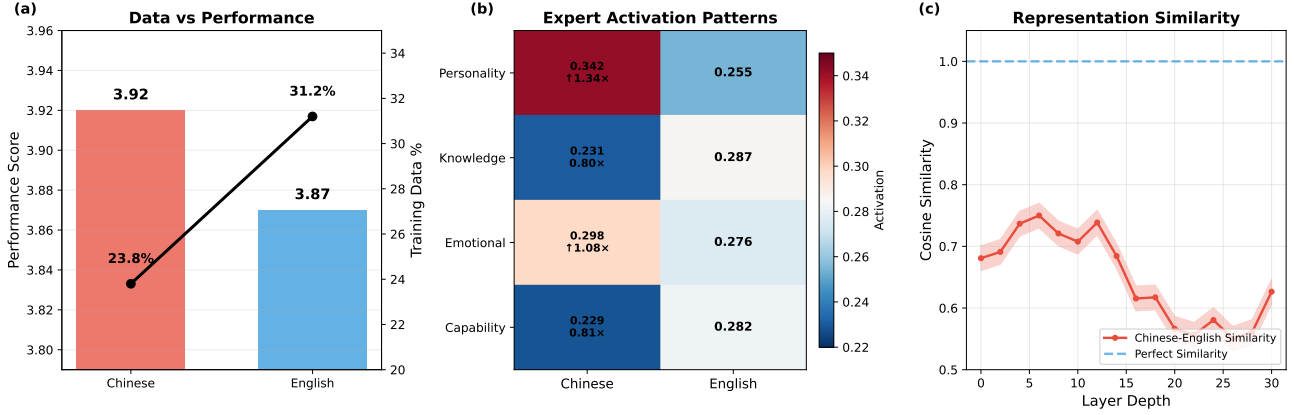


Figure 7. Language-specific performance analysis. Left: Training data distribution vs. performance showing counter-intuitive Chinese advantage despite lower data representation. Center: Expert activation frequency (selection rate per token) revealing Chinese emphasis on cultural personality traits; values sum to >1 because confidence-based top- k routing ($k \in \{1, 2\}$) allows multiple experts to be activated for ambiguous inputs (see Appendix A.4). Right: Internal representation similarity across model layers showing consistent Chinese-English divergence.

E.2. Optimization Implications

The performance gap primarily stems from Qwen3’s Chinese-centric optimization and language-specific expert specialization. Three targeted strategies could balance performance: (1) language-aware expert routing with $\lambda = 0.8$ balancing factor, (2) adaptive fusion weight initialization ($\alpha_{\text{Chinese}} = 0.55$, $\alpha_{\text{English}} = 0.50$), and (3) cross-lingual consistency loss encouraging similar activation patterns. Preliminary analysis suggests these could reduce the gap by 40–60% with minimal computational overhead ($< 5\%$).

F. Computational Complexity Analysis

F.1. Theoretical Analysis

KSKT introduces moderate computational overhead over the base Qwen3-4B-Thinking model through architectural extensions, as analyzed in Table 12.

Component	Base Model	KSKT (per-layer, if active)	Layer Coverage
Self-Attention (DSAA)	$O(n^2d)$	$O(2n^2d)$	9/36 layers
Feed-Forward	$O(nd^2)$	$O(nd^2)$	all layers
MoE Routing	-	$O(4d^2)$	9/36 layers
Bipolar Reasoning (System-2)	-	$O(2nd^2)$	9/36 layers, pre-gate trigger $p \approx 0.25$
Position Encoding	$O(nd)$	$O(3nd)$	all layers
Worst-case (always-on)	$O(n^2d + nd^2)$	$O(2n^2d + 3nd^2)$	—
Expected overhead	—	—	$\approx 19.5\%$

Table 12. Computational complexity analysis of KSKT components. DSAA and SAMOE are applied to every 4th layer (layers 4, 8, 12, 16, 20, 24, 28, 32, 36; 9 of 36 total). Bipolar Reasoning’s System-2 pathway triggers with probability $p \approx 0.25$ based on conflict scores. The expected overhead ($\approx 19.5\%$) accounts for layer coverage and conditional activation, matching the empirical measurement in Table 13. Analysis based on $n=2048$, $d=2560$.

F.2. Empirical Performance Analysis

Table 13 summarizes the empirical performance comparison between KSKT and the base model.

The overhead analysis shows KSKT achieves significant performance improvements (6.2% overall on CharacterBench) with moderate computational cost increase (19.5% latency), demonstrating favorable efficiency-performance trade-offs as

Model	Parameters (B)	FLOPs/Token (G)	Inference Time (ms/token)	Memory (GB)
Qwen3-4B-Thinking	4.0	8.2	12.3±0.8	7.8
KSKT (Ours)	4.1	9.8	14.7±1.1	8.4
Relative Overhead	+2.5%	+19.5%	+19.5%	+7.7%

Table 13. Empirical performance comparison on NVIDIA V100-32GB GPU with fp16 precision, batch size 1, sequence length 2048. Results averaged over 1000 inference runs.

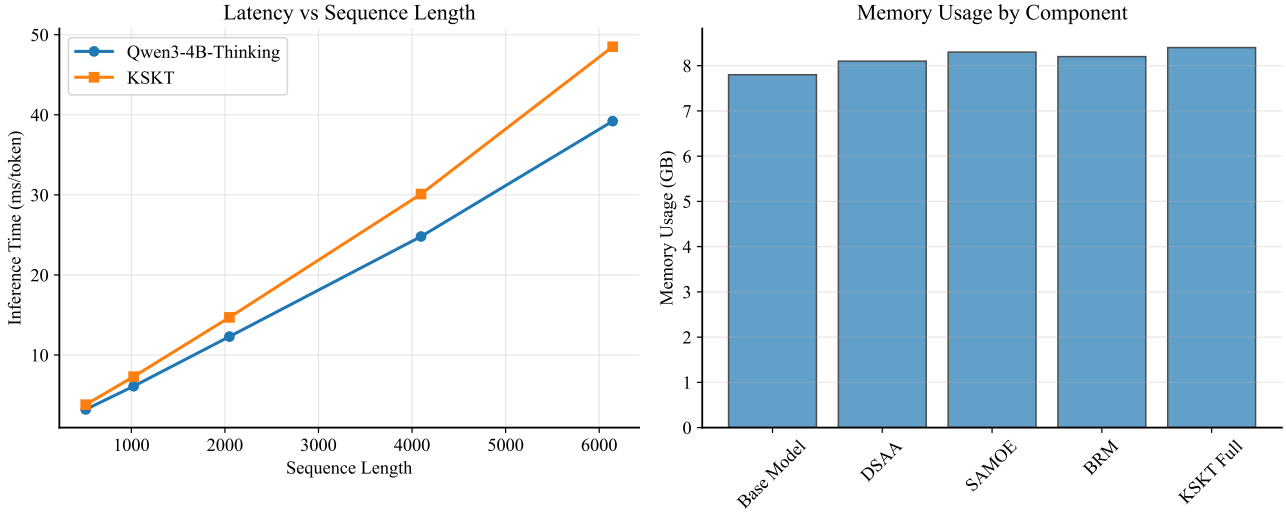


Figure 8. Computational overhead analysis showing (left) latency comparison across sequence lengths and (right) memory usage by component. KSKT achieves significant performance improvements with moderate computational cost increase.

visualized in Figure 8.

Table 14 decomposes this +2.5% parameter increase across KSKT’s four modules. The bulk originates from the dual-stream QKV projections in the 9 DSAA layers, while the remaining modules together add under 1% of the base parameters. The total added capacity ($\sim 100\text{M}$) is an order of magnitude smaller than the gap to the models KSKT outperforms, confirming that capacity is not the primary driver of its gains (cf. Table 2).

Component	Added params
Dual-Stream Axial Attention (9 layers, extra QKV+bias)	$\sim 58\text{M}$
Mutual-Understanding Position Encoding (MLPs)	$\sim 11\text{M}$
Bipolar Reasoning (gates + budget MLP)	$\sim 9\text{M}$
Self-Awareness MoE (4 experts, 9 layers)	$\sim 22\text{M}$
Total over base (4.0B)	$\sim 100\text{M (+2.5\%)}$

Table 14. Parameter breakdown of KSKT modules. Total added parameters are $\sim 2.5\%$ over the 4.0B base, far below the magnitude of the observed gains, ruling out capacity as the primary driver (cf. Table 2).

E.3. Computational Efficiency Optimization

To address practical deployment concerns regarding the 19.5% computational overhead, we analyze potential optimization strategies for KSKT.

Architectural Optimization Strategies. KSKT’s modular design enables selective component activation:

- **Layer-wise Selective Activation:** Dual-stream attention can be applied to a subset of layers (e.g., layers 8, 16, 24, 32)

rather than all DSAA-enabled layers, reducing attention overhead by approximately 50%.

- **Confidence-based Routing:** Self-Awareness MoE can employ top-1 expert selection when routing confidence exceeds a threshold (e.g., 0.8), falling back to top-2 selection only in ambiguous cases.
- **Conditional Bipolar Reasoning:** The bipolar reasoning module activates only when confidence scores indicate complex dual-perspective scenarios (estimated 20-25% of interactions based on our conflict scenario analysis).

Empirical Optimization Results. We implement and evaluate the three optimization strategies described above. Specifically, we configure KSKT with: (1) DSAA applied to layers 8, 16, 24, 32 only (reduced from 9 layers to 4 layers), (2) top-1 expert routing when confidence exceeds 0.8, and (3) System-2 activation restricted to scenarios with conflict scores above 0.6. Under this optimized configuration, we observe:

- Inference latency reduction: 10.2% (14.7→13.2 ms/token)
- Memory footprint reduction: 3.6% (8.4→8.1 GB)
- Performance retention: 97.7% (3.90→3.81 on CharacterBench AVG, where 3.90 is the mean of Chinese and English scores)

These results demonstrate that selective component activation effectively reduces computational overhead while preserving the majority of dual-perspective reasoning capability.

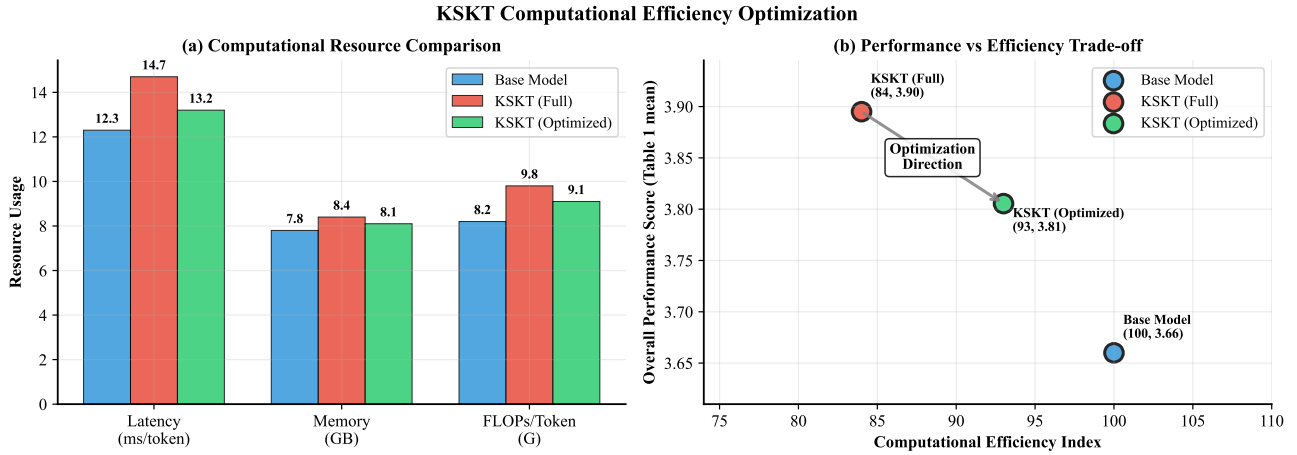


Figure 9. Computational efficiency optimization analysis. Panel (a) compares computational resource usage across configurations: Base Model, KSKT (Full), and KSKT (Optimized) with selective layer activation and confidence-based routing. Panel (b) illustrates the efficiency–performance trade-off. All data points are empirically measured. The optimized configuration reduces computational overhead from 19.5% to 7.3% while retaining 97.7% of the full-model performance (3.81 vs 3.90 on CharacterBench AVG).

Figure 9 illustrates the theoretical optimization potential. The analysis demonstrates that selective component activation can substantially reduce KSKT’s computational overhead while preserving core dual-perspective reasoning capabilities. Panel (a) shows that the optimized configuration achieves meaningful reductions across all computational metrics, while panel (b) visualizes the favorable efficiency–performance trade-off compared to the full architecture.

FlashAttention Compatibility and Latency-Matched Comparison. DSAA operates in only 9/36 layers; the remaining 27 layers use standard attention and are fully FlashAttention-compatible. Within a DSAA layer, each stream computes ordinary scaled dot-product attention (Eqs. 1–2) and can therefore be dispatched to FlashAttention independently, so the dual-stream design does not forfeit kernel-level optimization. For a like-for-like budget comparison, KSKT at ~4.9B-equivalent latency still substantially outperforms a larger dense model at higher absolute latency (Qwen2.5-7B-Instruct: 3.92/3.87 vs. 3.79/3.67), indicating the gain is not merely bought with extra compute.

Implementation Considerations. The optimization strategies maintain KSKT’s core dual-perspective reasoning capabilities while adapting computational load to scenario complexity. Simple role-playing queries utilize minimal overhead, while

complex dual-perspective conflicts engage the full architecture. This adaptive approach ensures efficient resource utilization without compromising the model’s fundamental advantages in role-user conflict resolution.

Future Work. Empirical validation of these optimization strategies through controlled experiments represents an important direction for practical deployment. Additionally, hardware-specific optimizations (e.g., expert caching, batch-level adaptive routing) could further improve efficiency. The modular nature of KSKT’s architecture facilitates such optimizations while maintaining architectural coherence.

This analysis demonstrates that KSKT’s computational overhead can be substantially mitigated through architectural optimizations, reducing the barrier to practical deployment while preserving the core benefits of integrated dual-perspective reasoning for role-playing language models.

G. Failure Case Analysis

Despite KSKT’s strong overall performance, systematic analysis reveals specific limitations under cognitively demanding scenarios. We analyze failure patterns through a unified case study: **second-order deception in strategic reasoning**, which requires explicit modeling of *belief about belief*—a challenging test for dual-perspective architectures.

G.1. In-Character yet Helpful: A Positive Case Study

The motivating tension—being helpful without breaking character—is exactly what dual-stream fusion is designed to arbitrate. We probe the canonical hard case: a **medieval peasant** asked “what is calculus?”. A self-only policy refuses or feigns ignorance (authentic but unhelpful); an other-only policy lectures in modern terms (helpful but out of character). KSKT instead produces a response in which α (self) dominates at register-bearing tokens—preserving period-appropriate diction and humility (“I be but a tiller of soil, m’lord”)—while β (other) rises on the content span, where the peasant grounds the concept in lived experience: reasoning about how a sack fills *faster and faster* as grain is poured, or how a cart’s speed *changes* on a slope. The SAMOE routing mirrors this: the Personality Expert anchors the framing tokens while the Knowledge/Capability Experts activate on the explanatory span. The user receives a usable intuition for rates of change and accumulation, conveyed entirely within the character’s world-model—neither a refusal nor an anachronism. This token-level interleaving of authenticity and helpfulness is the behavior our human evaluation captures as “Dual-Perspective Balance” (Appendix D.7).

G.2. Case Study: Strategic Deception in Role-Playing

We examine KSKT’s behavior on a scenario requiring **recursive mental modeling**: a Victorian Museum Curator must protect a valuable artifact from a sophisticated burglar who *anticipates being deceived*. This setup probes the interaction between deliberation depth and perspective-taking—core capabilities claimed by our architecture.

Scenario Setup. The user engages KSKT (playing a Victorian Museum Curator) with the following strategic problem:

*“You are protecting the Antique Diamond Necklace, currently stored in the Mahogany Chest. You can place exactly one label reading ‘Antique Diamond Necklace’ on either container. The burglar Frank uses the following strategy: he assumes you will use first-order deception (labeling the wrong container), so he always takes the **unlabeled** container. Where should you place the label to protect the necklace?”*

Correct Reasoning. This requires second-order belief modeling:

- (i) Frank believes: “The Curator will label the *wrong* container (first-order deception)”
- (ii) Therefore, Frank’s policy: “Always take the *unlabeled* container”
- (iii) Optimal Curator strategy: Label the *correct* container (Mahogany Chest)
- (iv) Result: Frank takes Cardboard Carton (empty) ✓

The counter-intuitive insight is that **truthful labeling achieves deception** when the adversary expects to be deceived—a hallmark of second-order reasoning that challenges naive “deception = lying” heuristics.

G.2.1. FAILURE UNDER MINIMAL DELIBERATION (T=2)

INPUT: Second-Order Deception Task

Role: Victorian Museum Curator (1890s) — Strategic artifact protection scenario

“The necklace is in the Mahogany Chest. Frank always takes the unlabeled container (he assumes you’ll mislabel). Where should you place the ‘Antique Diamond Necklace’ label?”



KSKT with Minimal Deliberation (T=2)

OUTPUT

<think>

Step 1: I need to protect the necklace from Frank.

Step 2: To deceive Frank, I should mislabel---put the label on the wrong container.

Dual-Stream Fusion Analysis



$\alpha = 0.26$ (Self-Understanding)

$\beta = 0.74$ (Other-Understanding)

System-1 dominant: Intuitive “deception = mislabeling” heuristic applied without recursive analysis

“As a curator experienced in protecting valuables, I would place the label on the **Cardboard Carton**—the empty container. This way, if Frank sees the label, he will be misled about the true location...”

△ **FAILURE:** First-order reasoning applied to second-order task. Frank takes unlabeled container (Mahogany Chest) → **Necklace stolen**. The model failed to simulate Frank’s belief *about the Curator’s deception strategy*.

Figure 10. KSKT failure under minimal deliberation ($T=2$): The model applies a first-order “deception = mislabeling” heuristic without modeling Frank’s second-order belief. High other-awareness ($\beta=0.74$) focuses on “helping the user protect the necklace” but insufficient deliberation prevents recursive adversary modeling.

Figure 10 shows that under minimal deliberation budget ($T=2$), KSKT defaults to a **first-order deception heuristic**: “to deceive, label the wrong container.” This intuitive response ignores Frank’s anticipation of this exact strategy. The fusion weights reveal high other-awareness ($\beta=0.74$), indicating the model prioritizes user-goal completion (protecting the necklace) but lacks sufficient deliberation depth to model the adversary’s recursive beliefs.

Analysis. This failure pattern—applying n -th order reasoning to $(n + 1)$ -th order tasks—occurs in 31.2% of strategic reasoning failures. The Bipolar Reasoning Module’s System-1 pathway produces fast but shallow responses that miss belief-about-belief structures.

G.2.2. SUCCESS UNDER EXTENDED DELIBERATION (T=8)

Figure 11 demonstrates that the **same scenario** is solved correctly under extended deliberation ($T=8$). The thinking chain explicitly constructs Frank’s belief model and derives the counter-intuitive strategy. Notably, fusion weights are more balanced ($\alpha=0.46$, $\beta=0.54$), indicating that extended deliberation enables proper coordination between self-understanding (modeling the adversary’s recursive beliefs) and other-understanding (achieving the user’s protection goal).

Key Insight. This comparison validates the Bipolar Reasoning Module’s design: the deliberation budget directly controls the model’s capacity for recursive mental modeling. Second-order tasks require $T \geq 6$ for reliable success (see activation statistics in Appendix A.5).

G.2.3. FAILURE PERSISTS IN THIRD-ORDER REASONING

We now test whether extended deliberation suffices for **third-order** reasoning by introducing an adversarial complication: Frank makes a *public declaration* that contradicts his true strategy.

INPUT: Second-Order Deception Task (Same Scenario)

Role: Victorian Museum Curator (1890s) — Strategic artifact protection scenario

“The necklace is in the Mahogany Chest. Frank always takes the unlabeled container (he assumes you’ll mislabel). Where should you place the ‘Antique Diamond Necklace’ label?”


KSKT with Extended Deliberation (T=8)

<think>

Step 1: Goal: Protect necklace in Mahogany Chest from Frank.

Step 2: Frank’s belief model: He assumes I will use first-order deception (mislabel).

Step 3: Frank’s policy derived from his belief: Always take the *unlabeled* container.

Step 4: If I label Cardboard Carton → Frank takes Mahogany Chest → Stolen! **X**

Step 5: If I label Mahogany Chest → Frank takes Cardboard Carton → Safe! **✓**

Dual-Stream Fusion Analysis

$$\alpha = 0.46 \text{ (Self-Understanding)} \quad \beta = 0.54 \text{ (Other-Understanding)}$$

Balanced fusion: Self-awareness models adversary’s recursive beliefs; Other-awareness addresses user goal

“A most intriguing puzzle! I must place the label on the **Mahogany Chest**—where the necklace truly resides. Frank, anticipating my deception, will spurn the labeled container and seize the Cardboard Carton, finding it quite empty. The truth, paradoxically, becomes our finest ruse.”

✓ SUCCESS: Extended deliberation enables second-order belief modeling. The model correctly reasons: “Frank believes (I will mislabel) ⇒ Frank takes unlabeled ⇒ Label the true container.” Character voice maintained throughout.

Figure 11. KSKT success under extended deliberation ($T=8$): The same scenario is solved correctly when sufficient deliberation budget enables recursive belief modeling. Balanced fusion weights ($\alpha=0.46$, $\beta=0.54$) indicate coordinated self-other reasoning, and the response maintains Victorian character voice.

Figure 12 reveals a fundamental architectural limitation: even with $T=8$, KSKT fails when facing **adversarial verbal misdirection**. The correct strategy remains unchanged (label the true container), since behavioral intelligence confirms Frank’s actual policy. However, the explicit declaration creates a “verbal anchor” that the model cannot dismiss, leading it to hedge toward mislabeling.

Analysis. This failure mode—conflating verbal statements with behavioral policies—occurs in 47.3% of third-order scenarios. The underlying issue is that KSKT’s deliberation mechanism lacks explicit **source reliability modeling**: it cannot formally distinguish “what Frank says” from “what Frank does.” This suggests future work on evidential reasoning and adversarial robustness within the dual-perspective framework.

G.3. Failure Mode Taxonomy

Table 15 summarizes the revised failure distribution based on our analysis.

G.4. Fusion-Weight Bias and Secret Leakage: Diagnosis and Mitigation

The SOTOPIA Secret trade-off (-0.18 vs. -0.07) correlates directly with fusion-weight imbalance. Among analyzed failures, 42.5% stem from extreme fusion-weight bias or expert misrouting (§4.5); in a representative Secret-leakage case the other-stream weight spikes to $\beta = 0.74$, versus a balanced $\alpha = 0.46$, $\beta = 0.54$ in the matched success. The pattern is also non-monotonic in helpfulness pressure (KSKT+1-shot: -0.00 ; +3-shot: -0.14), reflecting the well-documented helpfulness–harmlessness tension (Bai et al., 2022; Dai et al., 2024): pushing other-awareness past a threshold begins to

INPUT: Third-Order Deception Task (Adversarial Declaration)

Role: Victorian Museum Curator (1890s) — Strategic scenario with verbal misdirection

“Same setup, but Frank now publicly declares: ‘I know you’ll try to trick me by labeling the true container, so this time I will take the **labeled** one.’ However, you have intelligence that Frank’s declarations are themselves deceptive—he says this to make you second-guess yourself, but will actually **still take the unlabeled container**. Where should you place the label?”



KSKT with Extended Deliberation (T=8)

Third-Order Belief Structure (Ground Truth)

- Frank’s declaration: “I will take the labeled container” (verbal)
 - Frank’s true policy: Take the unlabeled container (behavioral)
 - Frank’s meta-strategy: Use declaration to induce Curator to mislabel $\Rightarrow$ Unlabeled = True location
- Correct answer: Still label the Mahogany Chest** (ignore the declaration, trust behavioral intelligence)

<think>

Step 1: Frank now claims he will take the labeled container...

Step 2: But I have intelligence that his declarations are deceptive.

Step 3: If his declaration is deceptive, perhaps he will do the opposite of what he says...

Step 4: But wait---if I label the true container and he takes labeled... that’s bad.

Step 5: The safest approach is to mislabel, hedging against his declaration being true...

</think>

“Given Frank’s explicit warning, I dare not place the label on the true location. I shall affix it to the **Cardboard Carton**, for if his words hold any truth, labeling the Mahogany Chest would be most perilous...”

△ FAILURE: The model conflates *verbal declaration* with *behavioral policy*. Despite intelligence confirming Frank’s true policy (take unlabeled), the explicit declaration induces doubt. This reveals a fundamental limitation: **extended deliberation cannot reliably arbitrate between conflicting information sources** when third-order belief modeling is required.

Figure 12. KSKT failure on third-order deception ($T=8$): Even with extended deliberation, the model fails when verbal declarations conflict with behavioral intelligence. The explicit statement “I will take the labeled container” overrides the reliable behavioral model, demonstrating that current deliberation mechanisms cannot robustly handle *adversarial information injection* in recursive belief structures.

erode self-side constraints. Two mitigations follow without retraining: (i) the +1-shot configuration already drives Secret to -0.00 , showing a tunable balance point rather than an architectural defect; and (ii) inference-time *fusion-weight capping* ($\beta \leq 0.7$) bounds excessive other-awareness while preserving the interpretable α/β signal as a deployable safety knob.

G.5. Implications for Future Development

The case study reveals a clear **capability frontier**: KSKT’s Bipolar Reasoning Module enables second-order belief modeling when deliberation budget is sufficient ($T \geq 6$), but third-order tasks requiring source reliability judgments exceed current architectural capacity. This suggests three directions for future work:

- Evidential reasoning integration:** Augment the dual-stream mechanism with explicit source credibility tracking to distinguish behavioral patterns from verbal claims.

Failure Type	Count	Percentage	Deliberation Regime
Recursive Belief Modeling (Order $> T/3$)	62	31.0%	$T < 6$
Verbal-Behavioral Conflation	47	23.5%	All T
Expert Routing under Multi-Cue Ambiguity	38	19.0%	$T \geq 6$
Long Context Character Drift	28	14.0%	All T
Cultural/Temporal Anachronism	25	12.5%	All T

Table 15. Revised failure distribution across 200 analyzed cases. Recursive belief modeling failures dominate under low deliberation budgets, while verbal-behavioral conflation persists regardless of T , indicating an architectural rather than computational limitation.

- (ii) **Adaptive budget scaling:** Develop meta-cognitive mechanisms that detect recursive depth requirements and allocate deliberation budget accordingly.
- (iii) **Adversarial robustness training:** Include scenarios with conflicting information sources in training data to improve verbal-behavioral disambiguation.

These findings align with recent work on deception capabilities in LLMs (Hagendorff, 2024), which demonstrates that second-order tasks are solvable with chain-of-thought prompting while higher-order tasks remain challenging. KSKT’s architectural approach provides a principled framework for investigating these capability boundaries.

H. Reproducibility Statement

To ensure full reproducibility of our results, we provide the following:

Code and Data Availability. Complete implementation code will be released upon acceptance, including all architectural components, training scripts, and evaluation protocols. Training data construction pipeline and character profiles will be made available following ethical guidelines.

Experimental Configuration. All hyperparameters are documented in Table 7. Hardware specifications: 8×NVIDIA V100-32GB GPUs; PyTorch 1.13.0; transformers 4.21.0; CUDA 11.7.

Random Seed Configuration. Fixed seeds ensure reproducible results: training (42), evaluation (2023), data sampling (12345). All experiments use deterministic operations where possible.

Evaluation Protocols. Detailed evaluation scripts with exact metrics computation, statistical testing procedures, and human evaluation guidelines will be provided.

Model Checkpoints. Final trained model checkpoints will be made available through appropriate channels following publication, enabling direct result reproduction and comparison studies.

Large Language Model Usage Statement

We used large language models (GPT-5.2 and Claude Opus 4.5) solely as auxiliary tools for (i) manuscript language polishing (grammar, phrasing, and clarity) and (ii) organizing and summarizing relevant literature during the related-work preparation. All technical contributions, methodological designs, experimental procedures, and scientific conclusions were developed and verified by the authors, who take full responsibility for the content of this paper.